

MSc Individual Research Project

Causal Structure Discovery via a Self-Implemented FCI⁺ Algorithm

A Case Study on the Tennessee STAR Project

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Abstract

This project independently implements paper-aligned FCI and FCI+, validates them against exact graphical oracles and established software, and separates algorithm fidelity from a robust finite-sample application workflow.

The scientific difficulty is that latent common causes and selection can make local adjacency conditioning insufficient, while standard FCI's Possible-D-SEP search can be expensive and sample conditional-independence errors can propagate into PAG endpoints. The implementation therefore provides both the historical search schedule of Spirtes et al. (2000) and Algorithm 2 of Claassen et al. (2013), whose $O(N^{2(k+2)})$ result is a conditional-independence query bound under a fixed true observed-MAG degree bound k , faithfulness, and an exact constant-time oracle.

Validation proceeds in layers: unit and property tests; exact MAG m-separation fixtures; the published Figure 4(b) MAG and exhibited separator $\{U, V, Z\}$; a 15-run, five-scenario known-truth benchmark; executed differential comparisons with causal-learn 0.1.4.7 and pcalg 2.7-12; and finite-sample order, bootstrap, sensitivity, and chronology audits. The endpoint-complete oracle targets are independently specified repository fixtures rather than all being PAGs printed in the sources.

On the committed known-truth benchmark, robust FCI+ has mean skeleton F1 0.985, semantic-edge F1 0.985, and endpoint accuracy 0.840, compared with 0.977, 0.943, and 0.737 for paper-profile FCI; its mean logical CI count is 402 rather than 450. Paper-profile FCI+ and the two executed external references have mean semantic-edge F1 0.951 under the recorded settings. These results describe the tested cases and policies; they do not establish general statistical dominance.

The separate Tennessee STAR application analyses 6,325 kindergarten-cohort students from the official 11,601-record dataset. Its strongest application evidence is the design-supported observed-arm contrast: small-class assignment has 13.90 points higher observed kindergarten achievement than regular-class assignment (school-cluster 95% interval 5.21 to 22.07), whereas the regular-with-aide contrast is near zero. Because this outcome subset is post-assignment selected and the original randomization blocks and compliance histories were not reconstructed, the contrast is consistent with a beneficial early effect rather than an identified selected-subset causal effect. Later grade-3 contrasts are complete-case results and weaker evidence. Across discovery methods, skeleton agreement is more defensible than several order-, threshold-, or chronology-sensitive endpoint marks; no PAG in this report identifies a unique DAG or estimates a treatment effect.

Executive Summary

Contribution. This project independently implements paper-aligned FCI and FCI+, validates them against exact graphical oracles and established software, and separates algorithm fidelity from a robust finite-sample application workflow. The reusable package and the STAR analysis remain separate layers: the former contains domain-neutral algorithms, while the latter supplies design, coding, selection, and interpretation decisions.

Source fidelity. Standard FCI is mapped to the 2000 book’s initial adjacency search, collider phase, two ordered Possible-D-SEP searches, reset, and historical orientation closure. Its theorem establishes a correct partially oriented inducing-path representation under correct CI information, but explicitly did not establish maximal informativeness. Complete modern PAG orientation is therefore attributed separately to Zhang (2008). FCI+ is mapped to the augmented skeleton, candidate D-sep-link recognition, recursive hierarchy, minimal separator, and candidate-revisit stages of Claassen et al. (2013, Algorithm 2, p. 178).

Validation outcome. Exact-oracle tests establish recovery of the independently specified targets on the encoded fixtures. For the published Figure 4(b) MAG, the source-verifiable claim is narrower: the false $X - Y$ link is removed with the exhibited separator $\{U, V, Z\}$ and FCI+ D-SEP provenance. On that run, FCI+ uses 63 logical CI queries compared with 102 for FCI. The endpoint-complete target is repository derived, and passing fixtures are not a proof for every faithful MAG.

Established-software outcome. A matched 15-run benchmark executes local paper FCI, local paper FCI+, local robust FCI+, causal-learn 0.1.4.7 FCI, and pcalg 2.7-12 fciPlus. Robust FCI+ improves the mean endpoint-oriented metrics on this benchmark at a higher query cost than paper FCI+, while paper FCI+, causal-learn, and pcalg agree in their aggregate semantic metrics under the recorded settings. Tetrad 7.6.10 is a documentation-only comparison because it was not executed. The local improvement claim is integrated auditability—paper-profile control, CI/orientation traces, separator-source provenance, D-SEP diagnostics, per-edge explanations, stability audits, and bundled artifacts—not universal superiority.

Independent implementation outcome. CRAN pcalg 2.7-12 under R 4.5.3 was run on the same encoded panels with a Python-equivalent G-square test. It is treated as a differential reference, not ground truth. The focused PAG is exactly identical across all three algorithms. In the attrition and longitudinal panels, pairwise skeleton Jaccard values range from 0.765 to 0.929, but exact endpoint agreement on shared edges is lower. Cyclic variable orders preserve every pcalg skeleton yet reproduce the original PAG endpoints in only 67–78% of orders.

Robust application outcome. The paper implementations remain unchanged for fidelity testing. A separate practical profile is used for empirical interpretation. Across all 26 cyclic column shifts it reproduces exact graph equality for every tested cyclic order within each panel, retains the main target adjacencies in every order, and replaces chronology-conflicting definite arrows with unresolved circle endpoints. This improves stability without claiming new causal identification.

STAR outcome. The official data were converted into three independent application panels:

1. an attrition panel containing 5,744 students and a grade-3 observation indicator;
2. a longitudinal panel containing 2,787 complete cases and both kindergarten and grade-3 achievement;
3. a focused complete-case panel containing 2,976 students, excluding kindergarten achievement from the discovery node set.

The strongest result does not come from a PAG endpoint. It is a design-supported observed-arm contrast: small classes have higher observed kindergarten scores than regular classes, whereas adding an aide to a regular class does not show a comparable advantage. The positive interval is consistent with a beneficial early effect, but post-assignment outcome selection prevents interpreting it as an identified effect for the selected subset. The later grade-3 contrast is also positive among observed students and requires still stronger missing-outcome qualifications.

Causal interpretation.

- The randomized design, together with the positive school-cluster interval, provides a design-supported observed-arm contrast consistent with a beneficial early effect. It does not by itself identify a causal effect within the post-assignment observed-outcome subset.
- There is no corresponding evidence that a regular class with an aide improves achievement relative to a regular class.
- Kindergarten achievement is one of the most consistently retained structural correlates of grade-3 achievement and grade-3 observation.
- The focused bidirected treatment-outcome edge is replicated exactly by all three algorithms, but is threshold-sensitive and cannot override knowledge of randomization.
- Cross-implementation skeleton consensus is more credible than endpoint marks that conflict with chronology or change under variable order.
- The robust application profile supports the same main adjacencies but leaves their directions unresolved; this is the preferred finite-sample reading of the discovery output.
- PAG endpoints should be read as equivalence-class constraints, not regression coefficients or direct effects.

Bottom line. The case study demonstrates practical use of the algorithm and reveals where selection and modelling choices matter. It does not convert exploratory causal discovery into a substitute for design-based effect estimation.

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Abbreviations and Notation

CI	Conditional independence.
DAG	Directed acyclic graph.
MAG	Maximal ancestral graph, representing observed-variable relations after latent variables or selection variables have been marginalised or conditioned upon.
PAG	Partial Ancestral Graph, representing the invariant features of a Markov equivalence class of MAGs.
FCI	Fast Causal Inference.
FCI⁺	Sparse FCI extension based on a logical characterisation of D-SEP links and hierarchical separating sets.
D-SEP	Claassen et al.’s named separator construct in a MAG/m-separation setting; it is distinct from informal lowercase d-separation in a DAG.
Logical CI query	One request made by search control flow. A cache hit is still a logical request but need not execute the underlying statistical calculation again.
Historical FCI output	The 2000 book’s partially oriented inducing-path graph; a modern maximally informative PAG additionally uses later ancestral-graph and orientation-completeness results.
STAR	Tennessee Student/Teacher Achievement Ratio experiment.
$X \perp Y \mid S$	X and Y are conditionally independent given conditioning set S .
$X \circ - \circ Y$	Neither PAG endpoint is identified.
$X \circ \rightarrow Y$	Y is not an ancestor of X , while the endpoint at X remains unresolved.
$X \leftrightarrow Y$	Arrowheads at both endpoints; compatible with latent confounding but not proof of one specific hidden cause.
$X \rightarrow Y$	An invariant ancestral relation in the PAG; not necessarily a direct causal effect.

Part I

Algorithm, Implementation, and Validation

Introduction

1.1 Motivation

Many empirical studies ask causal questions in settings where not every common cause has been measured. Family background, institutional selection, health, motivation, and administrative recording processes may all generate dependence that cannot be represented honestly by a causally sufficient DAG over the recorded columns. Fitting a directed graph while silently assuming causal sufficiency can turn uncertainty into unjustified arrows.

FCI was designed for this problem. Under causal Markov, faithfulness, and appropriate sampling assumptions, it uses conditional-independence information to learn a PAG that remains valid when latent confounders and selection variables may exist (Spirtes et al., 2000; Richardson and Spirtes, 2002). The price of this broader model class is additional search and greater endpoint ambiguity. In particular, standard FCI’s Possible-D-SEP refinement may test many separating sets that are not contained in local adjacency neighbourhoods.

FCI⁺ addresses the search problem for sparse causal graphs. Claassen et al. (2013) provide a logical characterisation of D-SEP links and a hierarchical procedure for finding the relevant separating sets. For a fixed maximum degree k , their result yields a polynomial worst-case bound in the number N of observed variables, commonly written

$$O\left(N^{2(k+2)}\right).$$

This is a sparsity-dependent guarantee, not a statement that every finite dataset will be easy or statistically well behaved.

1.2 Project contributions

The central contribution is stated as one auditable proposition:

Key result. This project independently implements paper-aligned FCI and FCI⁺, validates them against exact graphical oracles and established software, and separates algorithm fidelity from a robust finite-sample application workflow.

That proposition contains two distinct layers.

1. **Algorithm contribution.** A self-contained Python implementation of FCI and FCI⁺, with multiple CI tests, explicit PAG endpoint operations, paper-aligned profiles, practical finite-sample profiles, diagnostics, stability selection, type annotations, and release validation.
2. **Application contribution.** A separate Tennessee STAR case study that applies the public algorithm API without importing STAR-specific assumptions into the package. It uses randomized-arm contrasts as an external design-based reference and PAG learning as a structural sensitivity analysis.

This separation is central to the research design. An algorithm should remain reusable and domain-neutral; an application must state its own coding, selection, measurement, clustering, and interpretation decisions.

1.3 Research questions

The report addresses five questions:

- RQ1.** How do the original FCI and FCI+ sources define their targets, assumptions, search stages, and guarantee boundaries?
- RQ2.** Does the implementation map those stages to explicit code, public profiles, and tests?
- RQ3.** What is recovered under exact-oracle and finite-sample known-truth conditions?
- RQ4.** How does the package compare fairly with established FCI software in executed benchmarks and documented capabilities?
- RQ5.** Which STAR conclusions are design based, robust at the skeleton level, unresolved at the endpoint level, or merely exploratory?

Chapters 3–5 address theory and implementation; the validation and software chapters assess controlled evidence; and the later chapters form a separate STAR application. This ordering prevents a real-data pattern from being used as evidence that the algorithm matches its sources.

Causal-Graphical Background

2.1 Conditional independence and graphical assumptions

Let V be the set of observed variables, L latent variables, and S selection variables. Constraint-based discovery begins from statements of the form

$$X \perp Y \mid Z,$$

where $Z \subseteq V \setminus \{X, Y\}$. A causal graph and probability distribution are linked by two central assumptions.

Causal Markov condition. Graphical separation implies probabilistic conditional independence.

Faithfulness. Probabilistic conditional independences arise from graphical separation rather than exact cancellation of causal effects.

Under causal sufficiency these assumptions support DAG-based methods such as PC. When latent or selection variables are present, the appropriate observed-variable object is an ancestral graph or MAG (Richardson and Spirtes, 2002). Different MAGs may encode the same conditional-independence model, so the identifiable output is a PAG rather than one unique graph.

2.1.1 DAG, latent projection, MAG, and PAG targets

A DAG is a causal representation over a specified variable set. Marginalizing latent variables and conditioning on selection variables can induce directed, bidirected, and undirected relations among the observed nodes. An ancestral graph represents those relations without explicitly drawing the hidden or selection nodes; a maximal ancestral graph (MAG) makes every missing edge correspond to an m-separation. A PAG then records exactly the endpoint features shared by the MAGs in one observational Markov equivalence class (Richardson and Spirtes, 2002; Zhang, 2008).

M-separation generalizes d-separation to mixed ancestral graphs. It is the criterion used by the exact graphical oracle in this project. A sample Fisher-Z or G-square test, by contrast, is an error-prone statistical decision about a distribution. Oracle recovery evidence therefore isolates graph logic; it cannot be transferred directly to finite-sample accuracy.

2.1.2 Adjacency, ancestry, and intervention are different claims

An adjacency says that the measured variables were not separated by the tested sets under the recorded policy. A tail or arrowhead can constrain ancestry across the represented equivalence class. Neither fact alone identifies a direct structural coefficient, an adjustment set, or an interventional contrast. Selection variables can also induce undirected marks. These distinctions determine the evidence hierarchy used in the STAR application.

2.2 Meaning of PAG endpoints

Table 2.1: Interpretation of common PAG edge marks.

Mark	Interpretation
$X \circ - \circ Y$	The adjacency is retained, but neither ancestral endpoint is identified.
$X \circ -> Y$	The arrowhead at Y excludes Y as an ancestor of X ; the endpoint at X remains unresolved.
$X -> Y$	The tail-arrowhead pattern supports X as an ancestor of Y through some path in every represented MAG. It does not assert that the edge is a direct structural coefficient.
$X <-> Y$	Both endpoints are arrowheads. The pattern is compatible with latent confounding, but it does not identify a unique latent variable and can be distorted by selection or CI-test error.
$X - Y$	An undirected ancestral-graph edge, often associated with selection structure.

The distinction between ancestry and direct effect is especially important. FCI-family algorithms identify equivalence-class features. They do not estimate causal effect magnitudes, adjustment sets, or intervention distributions without additional analysis.

Standard FCI in Spirtes et al. (2000)

3.1 Historical target and assumptions

The Fast Causal Inference algorithm was developed for a problem that is strictly broader than causal discovery under causal sufficiency. Let O denote the measured variables and let the full acyclic causal graph also contain variables that are not in O . FCI reasons from conditional independences in the marginal distribution over O ; it does not require that all common causes of measured variables are themselves measured (Spirtes et al., 2000, Chapter 6, §§6.3 and 6.7, pp. 129–130 and 139).

The population argument rests on two assumptions about the full causal system. The Causal Markov condition links graphical separation to probabilistic independence, while Faithfulness excludes additional independences created by special parameter cancellations (Spirtes et al., 2000, Chapter 3, §§3.4.1 and 3.4.3, pp. 29–31). The algorithmic theorem additionally presumes that the conditional-independence decisions supplied to the search are correct. These premises matter operationally. A sample Fisher-Z or G-square decision is an estimate of the oracle answer; it is not the oracle assumed by the theorem.

The terminology also has a historical boundary. The second edition states its output as a *partially oriented inducing path graph*. The modern ancestral-graph and MAG/PAG formulation was developed subsequently (Richardson and Spirtes, 2002). This report uses PAG notation because it is the natural representation in the implementation, but it does not retroactively attribute every modern PAG result to the 2000 book. In particular, complete orientation of all invariant marks depends on the later rule set of Zhang (2008).

Interpretation caution. In this chapter, “book-aligned FCI” means the search and orientation schedule in Spirtes et al. (2000, Chapter 6, especially pp. 139–145). “Completed PAG” refers to the later MAG/PAG formalism and completeness results. The two descriptions are related, but they are not historically interchangeable.

3.2 Why adjacency-only conditioning is insufficient

A PC-style adjacency search removes $X - Y$ when it finds a set S among the current local neighbours such that $X \perp Y \mid S$. Under causal sufficiency, this local strategy has a strong justification. With latent variables, however, a pair may remain adjacent after local search even though it is separable over the observed variables. A necessary separator can contain a node that is no longer adjacent to either endpoint by the time the search reaches the relevant conditioning depth.

The book introduces D-SEP to characterize where an observed separator can reside in the underlying inducing-path structure (Spirtes et al., 2000, Chapter 6, §6.7, p. 141). D-SEP itself is not directly available from the partially learned graph. FCI therefore constructs the observable superset Possible-D-SEP and performs a second, non-local separation search. This second search is the central difference between standard FCI and a purely adjacency-based constraint procedure.

The cost is deliberate. Enlarging the candidate set protects against a false retained adjacency, but it can require many high-order CI queries. Thus FCI’s latent-variable protection and its computational burden arise from the same mechanism.

3.3 Source-located algorithm schedule

Table 3.1 separates the stages actually described in the book from later implementation choices. Page numbers are printed book pages, not PDF-image indices.

Table 3.1: Standard FCI stages and their primary-source locations.

Stage	Operation	Evidence carried forward	Source locator
1	Complete graph and PC-style adjacency search	Remove locally separable pairs and store $\text{Sepset}(X, Y)$.	Chapter 6, narrative pp. 142–143; FCI steps A–B, pp. 144–145.
2	Initial collider/arrow-head phase	For each unshielded $A - B - C$, orient arrowheads at B when $B \notin \text{Sepset}(A, C)$.	Causal Inference Algorithm step C, p. 139; reused by FCI, p. 145.
3	Possible-D-SEP construction	Build, separately from each ordered endpoint, a computable superset of possible non-local separator nodes.	Definition immediately before the Fast Causal Inference Algorithm, p. 144.
4	Non-local CI refinement	Search subsets from both ordered Possible-D-SEP pools; remove a remaining edge when a separating set is found and update the sepset record.	FCI step D, pp. 144–145.
5	Endpoint reset and collider reconstruction	Return retained edges to unresolved marks and reconstruct colliders from the final separator information.	FCI continuation, p. 145.
6	Orientation closure	Apply the original propagation and discriminating-path clauses repeatedly until no additional mark can be oriented.	Causal Inference Algorithm step D, pp. 139–140; invoked by FCI, p. 145.

The corresponding control flow can be expressed without importing later finite-sample policies:

Listing 3.1: Source-faithful control-flow summary of standard FCI.

```

def fci_2000(observed_variables, independence_oracle):
    graph = complete_circle_graph(observed_variables)
    graph, sepsets = pc_adjacency_search(graph, independence_oracle)

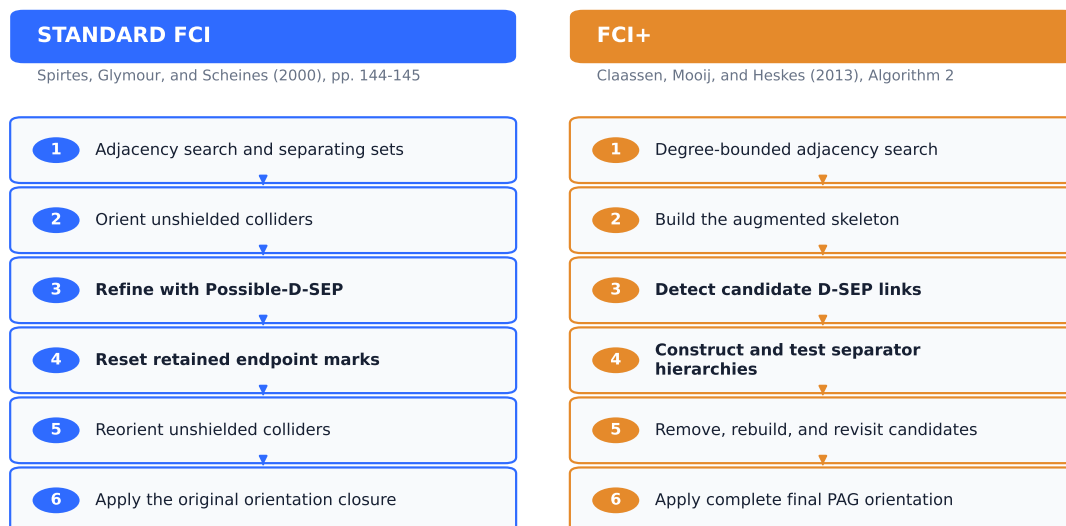
    orient_unshielded_colliders(graph, sepsets)
    search_graph = graph.copy()

    for x, y in search_graph.edges():
        pools = (
            possible_dsep(search_graph, x, y),
            possible_dsep(search_graph, y, x),
        )
        for pool in pools:
            # ordered directions stay separate
            separator = first_subset_separating(x, y, pool)
            if separator is not None:
                graph.remove_edge(x, y)
                sepsets[x, y] = separator
                break

    reset_endpoint_marks(graph)
    orient_unshielded_colliders(graph, sepsets)
    apply_original_orientation_closure(graph, sepsets)
    return graph

```

This listing is a report-level decomposition, not a quotation of the printed pseudocode. Its purpose is to make three easily missed obligations explicit: both endpoint orders are searched; the two candidate pools are not silently merged; and the final orientation phase is reconstructed after the additional edge removals.



Shared target: an observational equivalence-class representation; neither workflow identifies a unique DAG or a treatment effect.

Figure 3.1: Source-located control-flow comparison. Standard FCI performs a non-local Possible-D-SEP refinement; FCI+ instead iterates augmented-skeleton, candidate D-sep-link, hierarchy, and minimal-separator stages before invoking final orientation. The robust application profile changes finite-sample policy, not the definition of Algorithm 2.

3.4 Possible-D-SEP

For an ordered pair (A, B) , a node $V \neq A$ belongs to $\text{Possible-D-SEP}(A, B)$ when there is a path from A to V such that every internal triple on the path satisfies at least one of two conditions:

1. the middle node is a collider on that path; or
2. the middle node is not a definite noncollider and the two outer nodes are adjacent, so the triple forms a triangle.

This is the definition immediately preceding the Fast Causal Inference Algorithm in Spirtes et al. (2000, Chapter 6, §6.7, p. 144). The triangle clause is not a claim that the middle node is a true collider. It retains a node when the current partial marks do not yet justify excluding it. Possible-D-SEP is therefore intentionally permissive: it is a search-safe superset, not an estimate of a unique causal neighbourhood.

The asymmetry is also substantive. The source requires candidate search from the A -side and from the B -side. A valid separator may be available from one ordered construction but not the other. An implementation that uses only one direction, or that creates conditioning sets by mixing arbitrary members of the two pools, is no longer executing the same search contract.

In the repository, `discovery/pdsep.py::possible_dsep` traverses ordered edge states and applies the collider-or-triangle predicate in `_is_pds_step_allowed`. The traversal may conservatively include a node reachable by a qualifying walk even when a simple-path implementation would encounter it differently. Such over-inclusion spends additional CI queries but preserves the intended superset role. The corresponding regression suite checks endpoint exclusion, blocked paths, cycles, path limits, definite noncollider tails, both ordered directions, and the rule that the two pools must not be mixed.

3.5 Collider and discriminating-path information

For an unshielded triple $A - B - C$, separator membership distinguishes the collider and noncollider cases. If

$$B \notin \text{Sepset}(A, C),$$

the book's Causal Inference Algorithm orients arrowheads into B (Spirtes et al., 2000, step C, p. 139). FCI applies this decision once before Possible-D-SEP construction and again after the final skeleton is known. The second application is necessary because the non-local search can change both the skeleton and the stored separator evidence.

The later orientation phase also uses definite discriminating paths. For a discriminated triple, inclusion of the middle node in the relevant separator supports a noncollider conclusion, while exclusion supports a collider conclusion (Spirtes et al., 2000, definition and Figure 6.14, pp. 138–140). The source repeats the orientation clauses to closure. The modern labels R1–R10 should not be projected wholesale onto this passage: the original book supplies the historical arrowhead and discriminating-path phase, whereas later work supplies a complete PAG rule schedule.

3.6 Theorem scope and the historical completeness boundary

Theorem 6.4 states that, under Faithfulness and correct population-level independence decisions, FCI returns a partially oriented inducing path graph for the observed margin (Spirtes et al., 2000, p. 145). This is a correctness result for the historical target. It supports the following bounded statement:

Key result. Under the book’s graphical assumptions and correct CI decisions, the original FCI schedule returns a correct partially oriented inducing-path representation for the observed variables.

The sentence immediately following the theorem is equally important: the authors state that they did not know whether the output was always maximally informative. Therefore Theorem 6.4 does *not* prove orientation completeness for a modern completed PAG. The repository preserves this distinction with two orientation strategies:

- `orientation_strategy="spirtes_2000"` stops after the book-aligned historical closure; and
- `orientation_strategy="standard"` additionally applies the later Zhang-style tail-completion rules required for a complete modern PAG (Zhang, 2008).

The distinction is not cosmetic. A graph with more circles can be correct but less informative. Conversely, describing a later, more extensively oriented graph as the literal output theorem of the 2000 source overstates what that source established.

3.7 What is not established by the 2000 source

The following propositions are outside the theorem and algorithm statements in Spirtes et al. (2000):

1. **No finite-sample recovery guarantee.** The source guarantee is conditional on correct CI decisions. It does not prove that Fisher-Z, G-square, kernel CI, or a missing-data test will recover the oracle decisions at a given N .
2. **No universal tractability claim.** Possible-D-SEP may be large, and its subset search can remain expensive. The book discusses favourable sparse structures; it does not give the fixed-degree polynomial query bound later proved for FCI⁺.
3. **No 2000 proof of maximal orientation.** Maximal informativeness is explicitly left open after Theorem 6.4.
4. **No unique DAG.** The output represents observationally indistinguishable causal structures under the assumptions.
5. **No direct-effect or treatment-effect estimate.** A retained adjacency is not automatically a direct structural coefficient, and the algorithm does not estimate an intervention contrast.
6. **No identified latent variable from one edge mark.** A bidirected endpoint pattern is compatible with relevant ancestry restrictions, but it does not name, count, or measure a hidden cause.
7. **No guarantee under truncated search.** Finite conditioning or path limits, or disabling Possible-D-SEP, may leave a false adjacency and are not covered by the unbounded source schedule.

FCI+ in Claassen et al. (2013)

4.1 Problem statement and restricted complexity result

Claassen et al. (2013) ask whether causal structure discovery remains intrinsically exponential when the observed causal structure is sparse but latent variables and selection bias may be present. Their formal C-LEARN task assumes:

1. an independence oracle that answers any $X \perp Y \mid Z$ query exactly and in constant time;
2. a distribution faithful to an underlying causal DAG;
3. an observed-variable MAG whose maximum node degree is bounded by a constant k ; and
4. a completed PAG as the output target.

These assumptions appear in Definition 1 and the C-LEARN statement on p. 173, the input to Algorithm 2 on p. 178, and the complexity analysis on p. 179 of the UAI paper. Under them, the authors establish the worst-case conditional-independence query bound

$$O\left(N^{2(k+2)}\right),$$

where N is the number of observed variables and k is treated as a constant.

The claim is deliberately narrower than the paper’s memorable title. It is a polynomial query-complexity result for a sparse, faithful oracle problem. It neither solves an arbitrary NP-hard structure-learning problem nor makes a finite-sample statistical test exact or constant-time. The paper itself states that the result does not contradict known hardness results (Claassen et al., 2013, abstract and introduction, pp. 172–173).

4.2 D-separated pairs, D-sep nodes, and D-sep links

After a bounded PC adjacency search, some nonadjacent pairs in the true MAG can remain falsely connected because every useful separator requires a node outside the current endpoint-adjacency union. Definition 2 on p. 176 names such a pair D-separated and calls the surviving false adjacency a D-sep link (Claassen et al., 2013). A required nonadjacent member of the separator is a D-sep node.

The capitalization matters. D-SEP is the authors’ named search construct in a MAG setting. The paper’s surrounding preliminaries use m-separation for mixed ancestral graphs even though Definition 2 uses the phrase “d-separated.” This report therefore uses *D-sep link* for the named algorithmic object and *m-separation* for the exact graphical oracle.

Figure 4(b), pp. 175–176, gives the canonical five-node example. Its MAG has the six edges

$$Z \rightarrow U, \quad Z \rightarrow V, \quad U \leftrightarrow X, \quad V \rightarrow X, \quad U \rightarrow Y, \quad V \leftrightarrow Y.$$

The initial adjacency search retains a false $X - Y$ link, while

$$X \perp Y \mid \{U, V, Z\}$$

and Z is not adjacent to either endpoint. The source exhibits this separator and identifies Z as the D-sep node. It does not print the repository’s completed-PAG endpoint target, and it does not state that the exhibited separator is uniquely minimal.

4.3 The temporary augmented skeleton

FCI+ does not search every subset of standard FCI’s potentially large Possible-D-SEP pool. It first extracts additional invariant non-ancestry information into an *augmented skeleton* G^+ . The construction uses a single-node minimal-dependence test. If a known minimal independence

$$X \perp Y \mid S$$

becomes a dependence when a suitable adjacent node A is added,

$$X \not\perp Y \mid S \cup \{A\},$$

Lemma 2(1) supports an invariant arrowhead at A on the relevant incident edges. Definition 3 and the following paragraph on p. 176 define the augmentation; Algorithm 2 rebuilds it at lines 2 and 16 (Claassen et al., 2013).

The arrowheads in G^+ are search marks. They help recognize candidate D-sep links; they are not automatically final PAG orientations. The repository mirrors this distinction by copying the current skeleton in `build_augmented_skeleton`, using the copy during the hierarchical search, and resetting the surviving output skeleton before final collider and PAG orientation.

4.4 Recognizing candidate D-sep links

Lemma 4 on p. 176 supplies the characteristic augmented-skeleton pattern. Subject to the hierarchy condition developed in the paper, a D-sep link $X - Y$ appears as the middle edge of

$$U \leftrightarrow X \leftrightarrow Y \leftrightarrow V,$$

where U and V are nonadjacent and the required cross-direction paths do not run against an arrowhead. The Appendix A proof sketch on p. 180 explains why the local marks expose the otherwise non-local separator (Claassen et al., 2013).

This pattern only proposes a candidate. It is not itself a CI decision. FCI+ must still build a hierarchy-derived conditioning set and query the oracle. The distinction protects the soundness of the removal stage: arrowhead patterns identify where to search; independence evidence determines whether the edge is deleted.

The implementation keeps the pattern literal. The middle and witness edges must be bidirected in the augmented graph; unresolved circle marks are not accepted as substitutes for invariant arrowheads. `_has_dsep_link_witness` also checks witness nonadjacency and the two not-against-arrowhead reachability conditions. The associated tests include positive witnesses, missing cross paths, missing target arrowheads, and circle-only negative cases.

4.5 Recursive hierarchy and removal order

Definitions 4–5 and Lemmas 6–7 on p. 177 define the hierarchy $\text{HIE}(X, I)$. Starting from a seed set, it repeatedly adds every node that occurs in a recorded minimal separator between a pair of nodes already in the hierarchy. In fixed-point form,

$$\begin{aligned} H_0 &= X, \\ H_{t+1} &= H_t \cup \bigcup_{\substack{A, B \in H_t \\ A \neq B}} \text{Sepset}_I(A, B), \\ \text{HIE}(X, I) &= H_t \quad \text{when } H_{t+1} = H_t. \end{aligned}$$

Here I is the paper’s collection of minimal independence statements. Lemma 6 explains why a required D-sep node appears in another relevant minimal separator, while Lemma 7 shows how the hierarchy seeded by suitable adjacent ancestors contains a separator for the target D-sep link (Claassen et al., 2013, pp. 177–178).

Candidate D-sep links can depend on one another. Lemma 5 gives an acyclic ordering argument: removal of one ancestral D-sep link may reveal the information required to remove another, but the dependencies need not be solved simultaneously. Section 4.3 and Algorithm 2 implement this reasoning by rebuilding the augmented skeleton and reconsidering candidates after each successful deletion.

The repository function `hierarchy` performs this fixed-point expansion and explicitly excludes the target pair’s own separator from recursive lookup. The D-SEP refinement caches expansions by both the seed set and the excluded target pair. That cache is an engineering optimization; the hierarchy definition and candidate-revisit logic are the paper mechanisms.

4.6 Algorithm 2

Table 4.1 maps the printed lines of Algorithm 2 to their logical role. The source location is Claassen et al. (2013, §5, pp. 178–179).

Table 4.1: FCI+ Algorithm 2: source-located stages.

Algorithm 2 lines	Stage	Function in the proof and search
1	Bounded PC adjacency search	Construct the initial skeleton G and minimal-independence set I using conditioning depth bounded by k .
2–3	Augment and propose	Construct G^+ from single-node dependencies and obtain candidate D-sep links from its invariant arrowhead patterns.
4–12	Enumerate sparse bases	For a candidate $X - Y$, enumerate at most k -node endpoint bases and seed the recursive hierarchy from $\{X, Y\}$ and the selected bases.
13–15	Test and minimize	Query whether the hierarchy separates X and Y ; if so, remove redundant members and add a minimal separator to I .
16–18	Rebuild and revisit	Remove the confirmed D-sep link, rebuild G^+ , and reconsider candidate links using the expanded independence information.
23–24	Complete orientation	Orient unshielded colliders and invoke the complete FCI/PAG rule machinery to return the completed PAG.

A compact implementation-neutral transcription is:

Listing 4.1: Control-flow summary of Claassen et al.’s Algorithm 2.

```

def fci_plus(observed_variables, independence_oracle, k):
    graph, independence_set = pc_adjacency_search(
        observed_variables, independence_oracle, depth=k
    )
    augmented = augment(graph, independence_set, independence_oracle)
    candidates = possible_dsep_links(augmented)

    while candidates:
        x, y = candidates.pop()
        for zx in endpoint_bases(augmented, x, exclude=y, max_size=k):
            for zy in endpoint_bases(augmented, y, exclude=x, max_size=k):
                seed = {x, y, *zx, *zy}
                h = hierarchy(seed, independence_set, exclude_pair=(x, y))
                if independence_oracle.independent(x, y, h - {x, y}):
                    separator = minimize(h - {x, y})
                    graph.remove_edge(x, y)
                    independence_set[x, y] = separator
                    augmented = augment(
                        graph, independence_set, independence_oracle
                    )
                    candidates = possible_dsep_links(augmented)
                break

    reset_endpoint_marks(graph)
    orient_colliders(graph, independence_set)

```

```

apply_complete_pag_orientation(graph, independence_set)
return graph

```

The listing emphasizes two properties sometimes lost in simplified descriptions. First, k bounds the endpoint bases because it is assumed to bound the true observed-MAG degree; it is not merely a convenient runtime knob. Second, a successful edge removal changes the independence information and therefore requires augmentation and candidate detection to be repeated.

4.7 Query-complexity accounting

The main paper’s p. 179 accounting separates the stages:

- the bounded PC adjacency search uses $O(N^{k+2})$ CI queries;
- construction of the augmented skeleton uses $O(N^3)$ queries;
- at most $O(N^2)$ false-edge candidates are considered;
- the paired endpoint bases contribute $O(N^{2k})$ choices for a candidate;
- separator minimization is separately bounded by $O(N^2)$ queries, while repeated candidate reconsideration supplies an additional $O(N^2)$ factor in the worst-case dominant accounting; and
- final graph orientation uses no additional CI queries, although it has its own graph-operation cost.

Consequently the dominant CI-query count is

$$O(N^2) O(N^{2k}) O(N^2) = O(N^{2(k+2)}).$$

This is the theorem/body value used throughout the report. The arXiv proof supplement’s abstract states $N^{2(k+1)}$, but its Theorem 1, proof, and concluding sentence agree with the main UAI paper on $N^{2(k+2)}$. The discrepancy is recorded rather than silently harmonized.

Logical query count must be distinguished from elapsed time. The theorem treats each oracle answer as a constant-time operation. A sample CI test may fit regressions, form contingency tables, perform matrix operations, or run permutations; its cost can depend on sample size and conditioning-set dimension. Likewise, repository caching can reduce underlying test executions without changing how often algorithmic control flow requests a query.

4.8 Orientation completeness is a dependency

FCI+’s novel contribution is the sparse recovery of the correct skeleton and separating information. Algorithm 2 line 23 then invokes complete FCI orientation. The UAI paper does not rederive the full orientation-completeness proof; it relies on established PAG orientation results and cites Zhang (2008). Therefore the complete-output claim requires all of the following:

1. correct CI decisions and separators;

2. correct identification and removal of every D-sep link;
3. sound collider decisions; and
4. a complete invariant-mark orientation schedule.

In the repository, the paper profile selects `orientation_strategy="standard"`, which applies the later complete schedule. The historical "`spirtes_2000`" schedule and the "`conservative`", "`leaf`", and "`robust`" engineering strategies intentionally have different orientation scopes and must not inherit the complete-PAG theorem by label alone.

4.9 What is not established by the 2013 source

1. **No finite-sample superiority theorem.** The paper does not prove that FCI+ is more accurate than standard FCI for a fixed sample, CI test, or significance threshold.
2. **No empirical constant-time oracle.** A statistical CI test is neither exact nor necessarily constant-time.
3. **No polynomial guarantee for growing or wrong k .** The theorem assumes a constant bound on the true observed-MAG degree. A user-entered `sparsity_bound` is not evidence that the assumption is true.
4. **No unrestricted complexity-class conclusion.** The result is for the restricted sparse oracle problem, not every causal discovery formulation.
5. **No wall-clock dominance.** Query complexity does not rank programming languages, CI implementations, memory use, or practical runtimes.
6. **No claim that every hierarchy is statistically efficient.** The discussion on p. 179 warns that hierarchy-derived conditioning sets may be large and may reduce finite-sample power.
7. **No unique causal graph or effect estimate.** The target is a completed PAG equivalence class, not a unique DAG or an intervention distribution.
8. **No published completed PAG in Figure 4(b).** The paper prints the MAG example and an exhibited separator. A completed endpoint target must be derived and validated separately.

Independent Python Implementation

5.1 Design objective and package boundary

The implementation objective is not to relabel a generic constraint-based search as FCI+. It is to expose the algorithmic obligations identified in Chapters 3 and 4 as inspectable Python components, and to keep them separate from finite-sample application policy. The primary contribution can be stated in one sentence:

Key result. The repository independently implements paper-aligned standard FCI and FCI+ search profiles, connects each major source concept to a named code symbol and regression test, and adds a separately labelled auditability and finite-sample robustness layer.

The reusable code resides in `src/fci_engine`; the Tennessee STAR workflow resides in `case_studies/tennessee_star`. No STAR variable, temporal constraint, discretization choice, or substantive conclusion is embedded in the estimator package. This boundary makes the case study an application of the public API rather than an implicit modification of the algorithm.

Table 5.1: Implementation layers and responsibilities.

Layer	Responsibility	Principal location
Graph representation	Store one endpoint mark at each side of every adjacency; validate mutation, query ancestry, and serialize semantic edge records.	graph/endpoint.py; graph/pag.py
CI abstraction	Provide one test protocol, concrete continuous/discrete/missing/nonlinear tests, canonical query caching, and event-level provenance.	ci/
Shared discovery lifecycle	Validate data, resolve profiles, build the initial skeleton, orient, apply background knowledge, and assemble results consistently.	discovery/base.py
Standard FCI	Run the initial skeleton, first collider phase, ordered Possible-D-SEP refinement, reset, and historical or modern orientation closure.	discovery/fci.py; discovery/pdsep.py
FCI+	Run bounded adjacency search, augmented-skeleton candidate discovery, hierarchical D-SEP removal, reset, and complete or conservative orientation.	discovery/ fci_plus.py; discovery/dsep.py
Validation and audit	Supply exact MAG m-separation fixtures, endpoint metrics, stability wrappers, reference comparisons, explanations, and artifact export.	simulation/ metrics/ result.py

5.2 Public API and configuration

The top-level functions keep standard FCI and FCI+ as separate entry points. Paper profiles are explicit because a default optimized for finite samples should not silently stand in for the source algorithm.

Listing 5.1: Paper-aligned entry points remain distinct.

```

from fci_engine import fci, fci_plus
from fci_engine.config import FCISConfig, FCISPlusConfig

fci_result = fci(
    data,
    config=FCISConfig.paper(),
)

fci_plus_result = fci_plus(
    data,
    config=FCISPlusConfig.paper(k=3),
)

```

`FCISConfig.paper()` fixes the unbounded book-aligned search choices: immediate skeleton updates, first-found separators, enabled Possible-D-SEP, no conditioning or path cap, nonconservative collider ori-

entation, and `orientation_strategy="spirtes_2000"`. `FCIPlusConfig.paper(k=...)` requires an explicit k , assigns that same value to initial conditioning depth and the endpoint-base sparsity bound, disables ordinary Possible-D-SEP, uses immediate deletion and first-found separators, and selects the later complete "standard" orientation schedule required by Algorithm 2's output claim.

The profiles constrain execution; they do not validate theorem assumptions. Supplying $k = 3$ does not prove that the unknown true MAG has degree at most three. Likewise, using a paper profile with the default Fisher-Z test reproduces the search schedule but does not create the constant-time exact oracle assumed in the theory.

5.3 PAG endpoint representation and mutation invariants

The PAG class in `graph/pag.py` stores endpoints for ordered node pairs while exposing each adjacency once. Its endpoint accessor returns the mark at y on edge $x - y$. This convention permits explicit representation of tails, arrowheads, circles, and absence without flattening a PAG into a directed graph.

Mutation operations validate both nodes, reject a one-sided or NONE-marked edge, and require edge deletion through `remove_edge`. Query methods distinguish adjacency, possibly-directed ancestry, and directed-path ancestry. These invariants matter because an endpoint error can change both Possible-D-SEP reachability and later orientation rules.

Tests in `tests/test_pag.py` exercise construction, node validation, edge addition/removal, endpoint mutation, edge rendering, and copy independence. `tests/test_pag_queries.py` separately tests possible- and definite-ancestor semantics. The implementation therefore treats graph storage as a tested algorithmic substrate, not a plotting-only convenience.

5.4 Conditional-independence protocol, caching, and provenance

Every search uses the abstract `CITest.test(data, x, y, cond_set)` contract and receives a structured `CITestResult`. The built-in tests cover Fisher-Z, missing-value Fisher-Z, chi-square, G-square, and kernel CI. A custom test can implement the same protocol, which keeps search logic independent of a particular data type.

`CITestCache` canonicalizes the unordered endpoint pair and conditioning set. Every request increments the logical query count and emits a `CITraceEvent`; a repeated request also records that the result was a cache hit. Thus the result distinguishes:

- the number of logical CI requests made by algorithmic control flow;
- the subset answered from the cache; and
- the underlying statistical test results, including method, statistic, p -value, effective sample size, and conditioning set.

This distinction is important when connecting measurements to the FCI+ theorem. The theorem counts oracle queries. Repository elapsed time includes data preparation, graph operations, and the cost of uncached sample tests. Neither count should be silently substituted for the other.

5.5 Standard FCI pipeline

`discovery.fci.FCI.fit` exposes the source stage ordering directly:

1. `_learn_initial_skeleton` creates a complete circle graph and calls `learn_initial_skeleton`;
2. `_orient_colliders` applies separator-based unshielded-collider decisions;
3. `refine_skeleton_with_pdsep` searches subsets from the two ordered Possible-D-SEP pools and records removals with source "pdsep";
4. `_orient_colliders(..., reset=True)` clears provisional endpoint marks and reconstructs collider information from the final sepsets; and
5. `_apply_orientation_rules` runs the configured historical, complete, or conservative closure.

The Possible-D-SEP traversal uses ordered edge states rather than enumerating all complete simple paths. Its step predicate accepts a collider or a triangle whose center is not proven a noncollider by a tail. The compact representation does not separately encode the book's underlined-noncollider notation. It may conservatively over-include walk-reachable nodes; this is compatible with the superset role but can increase CI work. Dedicated tests retain coverage of path blocking, cycles, tail-proven noncolliders, ordered endpoint pools, and stable versus immediate update schedules.

5.6 FCI+ pipeline

`discovery.fci_plus.FCIPlus.fit` deliberately does not call standard FCI's Possible-D-SEP refinement. It passes the initial skeleton and separating sets to `refine_skeleton_with_fci_plus_dsep`, which implements the Algorithm 2 search:

1. `build_augmented_skeleton` copies the skeleton and adds single-node-dependence arrowheads;
2. `possible_dsep_links` recognizes literal bidirected witness patterns and cross-direction paths;
3. `_algorithm2_base_sizes` and `_base_combinations_for_sizes` enumerate the paired endpoint bases;
4. `hierarchy` expands each seed through recorded separators to a fixed point;
5. the CI interface tests the resulting conditioning set;
6. `minimal_dsep` removes redundant separator members;
7. the edge and augmented copy are updated, the source "fci_plus_dsep" is recorded, and candidates are rebuilt; and
8. the final estimator resets the temporary marks and applies collider plus configured PAG orientation.

`DSEPDiagnostics` records candidate sightings and revisits, hierarchy queries and cache hits, duplicate conditioning-set skips, D-SEP-stage CI requests, removed edges, and the maximum tested conditioning size.

These counters make an Algorithm 2 run inspectable. They are empirical accounting, not a mechanized proof of the asymptotic bound.

The Figure 4(b) regression has a narrow source-grounded target. The repository encodes the six published MAG edges, verifies by exact m-separation that $\{U, V, Z\}$ separates X and Y , and checks that the end-to-end FCI+ run removes the false adjacency in the hierarchical stage with the "fci_plus_dsep" provenance tag. The completed endpoint fixture is repository-derived and is not described as a PAG printed in the paper.

5.7 Orientation engine and strategy separation

`discovery.rules.apply_orientation_rules` provides one dispatcher with explicit schedules. The separation is part of the scientific interface:

Table 5.2: Orientation strategies and the claims they can support.

Strategy	Intended role	Claim boundary
<code>spirtes_2000</code>	Original arrowhead and discriminating-path closure used by the standard-FCI paper profile.	Historically aligned; not claimed maximally informative.
<code>standard</code>	Modern R1–R10-style complete schedule, including later tail-completion rules.	Appropriate completeness dependency for the FCI+ paper profile under its oracle assumptions.
<code>conservative</code>	Retain arrowhead conclusions while suppressing tail-producing propagation.	Finite-sample engineering policy; no completed-PAG claim.
<code>leaf</code>	Permit a restricted R1 tail when the newly directed endpoint is a leaf.	Engineering compromise between informativeness and aggressive propagation.
<code>robust</code>	Combine conservative collider decisions with the leaf-tail schedule.	Application profile for the specified stability audits, not a paper algorithm.

Rule-level tests cover the original propagation and discriminating-path patterns, closure behavior, contradiction avoidance, and the later R5–R10 tail rules. A separate published-reference regression verifies that the `spirtes_2000` paper profile stops before a selection-bias tail completion that the later `standard` schedule can orient. This test makes the historical completeness boundary executable rather than merely descriptive.

5.8 Paper profiles and finite-sample engineering profiles

The repository separates algorithm fidelity from application robustness. The distinction is summarized in Table 5.3.

Table 5.3: Search profiles: source contracts versus engineering policy.

Profile	Search choices	Orientation	Permitted interpretation
FCI paper	Unbounded depth/path; immediate skeleton deletion; first separator; ordered Possible-D-SEP search.	<code>spirtes_2000</code>	Closest executable mapping to the 2000 schedule; historical target is not claimed maximally informative.
FCI ⁺ paper	Explicit shared k ; immediate deletion; first separator; hierarchical D-SEP; no standard Possible-D-SEP.	<code>standard</code>	Closest mapping to Algorithm 2; theorem wording requires exact CI, faithfulness, and correct true-degree bound.
FCI practical/default	Stable skeleton and Possible-D-SEP; max- p separator selection; optional bounds.	<code>standard</code> unless overridden	Finite-sample implementation policy; any caps weaken source guarantees.
FCI ⁺ practical	Automatic- α heuristic; default depth and sparsity bounds of three; stable skeleton; max- p separators; no standard Possible-D-SEP.	<code>robust</code> with conservative colliders	Engineering extension assessed by finite-sample, order, bootstrap, and chronology audits; not Algorithm 2 verbatim.

The practical profile does not claim that uncertainty has been eliminated. Stable deletion reduces one source of variable-order dependence. Max- p separator selection spends more CI queries to retain stronger sample independence evidence at a fixed depth. Conservative collider and tail policies preserve circles where sample evidence is ambiguous. Automatic α is a documented sample-size heuristic. Each choice can improve a predefined robustness metric on a given benchmark while weakening informativeness or increasing computation.

The committed known-truth evaluation and STAR order audit therefore compare profiles under their actual names. “Paper FCI+” is the fidelity and sparse query-search subject. “Robust FCI+” is the preferred engineering policy for the reported finite-sample application because it meets the stated audit criteria while returning less-specific endpoints. That preference is local to the evidence and criteria in this project; it is not a universal accuracy ranking.

5.9 Result object, explanations, and reproducible artifacts

`FCIResult` binds the final graph to the evidence that produced it. It stores the resolved configuration, sample and node counts, separating sets and their stage sources, logical CI count, cache hits, named CI event trace, orientation event trace, ambiguous triples, D-SEP diagnostics, elapsed time, and optional bootstrap adjacency frequencies.

`explain_edge(x, y)` returns a pair-specific record containing the final endpoint marks when present, the separator and its provenance when removed, all direct CI events for the pair, and orientation events attached to that edge. `save_artifacts` exports structured JSON, a semantic endpoint CSV, and standalone HTML. This design prevents a plotted arrow from becoming detached from the profile, CI test, threshold, or orientation decision that created it.

The package is typed and includes `py.typed`. The repository’s quality workflow runs Pytest, Ruff, strict MyPy, wheel construction, installation, and an import smoke test. These checks establish interface and packaging consistency. They do not validate faithfulness, true sparsity, CI-test model assumptions, or an empirical causal conclusion.

5.10 Source-to-code-to-test traceability

Table 5.4 is the core audit table. A classification of “paper mechanism” means that the symbol implements the stated source operation when invoked through the corresponding paper profile. It does not mean that the source proves every line of Python correct.

Table 5.4: Published concept to repository symbol and validation mapping.

Published concept	Repository symbol	Principal validation	Classification
Complete start and PC-style adjacency search	<code>discovery.skeleton.create_complete_pag;</code> <code>learn_initial_skeleton</code>	skeleton tests; profile and published-reference integration tests	Shared paper mechanism
Standard FCI stage ordering	<code>discovery.fci.FCI.fit</code>	<code>tests/test_published_reference_graphs.py</code>	Spirtes-aligned
Possible-D-SEP path criterion	<code>discovery.pdsep.possible_dsep;</code> <code>_is_pds_step_allowed</code>	<code>tests/test_pdsep.py</code> : collider/triangle, tail blocking, cycles, and path bounds	Spirtes-aligned
Ordered Possible-D-SEP separator search	<code>discovery.pdsep.refine_skeleton_with_pdsep</code>	both-direction and no-mixed-pool tests in <code>test_pdsep.py</code>	Spirtes-aligned
Reset and second collider phase	<code>FCI.fit;</code> <code>orientation.reset_endpoint_marks</code>	endpoint-reset unit test and published-reference regression	Spirtes-aligned
Original orientation closure	<code>rules.apply_orientation_rules(.., "spirtes_2000")</code>	orientation-rule tests; historical stopping regression	2000-aligned, incomplete by source
Complete modern orientation	<code>rules.apply_orientation_rules(.., "standard")</code>	<code>tests/test_oracle_pag_rules.py</code> ; schedule tests	Later Zhang dependency

Published concept	Repository symbol	Principal validation	Classification
Augmented skeleton	<code>dsep.build_augmented_skeleton;</code> <code>_augment_with_single_node_dependencies</code>	single-node-dependence and existing-edge tests in <code>tests/test_fci_plus.py</code>	Claassen Algorithm 2
Candidate D-sep pattern	<code>dsep.possible_dsep_links;</code> <code>_has_dsep_link_witness</code>	bidirected witness, cross-path, target-arrowhead, and circle-rejection tests	Claassen Algorithm 2
Recursive hierarchy	<code>dsep.hierarchy;</code> <code>_hierarchy_cache_key</code>	fixed-point expansion and excluded-pair cache-key tests	Claassen Algorithm 2
Paired endpoint bases bounded by k	<code>_algorithm2_base_sizes;</code> <code>_base_combinations_for_sizes</code>	separate-base and literal nested-order tests	Claassen Algorithm 2
Separator minimization and candidate revisit	<code>minimal_dsep;</code> <code>refine_skeleton_with_fci_plus_dsep</code>	fixed-point minimization, D-SEP removal, and failed-candidate revisit tests	Claassen Algorithm 2
Figure 4(b) false-link removal	<code>simulation.mag_oracle.canonical_dsep_mag;</code> <code>MAGOracleCITest</code>	direct $\{U, V, Z\}$ m-separation and <code>fci_plus_dsep</code> provenance regression	Source MAG plus repository-derived target
Exact MAG m-separation oracle	<code>simulation.mag_oracle.MAGSpec;</code> <code>MAGOracleCITest</code>	<code>tests/test_oracle_graphs.py;</code> <code>test_published_reference_graphs.py</code>	Validation apparatus
Stable deletion and max- p separators	<code>FCIConfig.practical;</code> <code>FCIPlusConfig.practical</code>	configuration, order, and finite-sample benchmark tests	Engineering extension
Conservative and robust orientation	<code>apply_orientation_rules;</code> <code>orientation_strategy</code>	<code>tests/test_orientation_rules.py;</code> <code>test_robust_application_profile.py</code>	Engineering extension
Structured trace and exports	<code>result.FCIResult;</code> <code>diagnostics.py</code>	<code>tests/test_result_exports.py</code>	Engineering extension

LITERATURE CONCEPT	IMPLEMENTATION SYMBOL	EXECUTABLE VALIDATION
SOURCE-ALIGNED Standard FCI schedule Spirtes et al. (2000), pp. 144-145	→ <code>discovery/fci.py::FCI.fit</code>	→ <code>test_published_reference_graphs.py</code>
SOURCE-ALIGNED Possible-D-SEP refinement Spirtes et al. (2000), p. 144	→ <code>discovery/pdsep.py::possible_dsep_refine_skeleton_with_pdsep</code>	→ <code>test_pdsep.py</code>
SOURCE-ALIGNED Augmented skeleton Claassen et al. (2013), pp. 176, 178	→ <code>discovery/dsep.py::build_augmented_skeleton_augment_with_single_node_dependencies</code>	→ <code>test_fci_plus.py::test_augmented_skeleton_orients_single_node_dependency_arrowheads</code>
SOURCE-ALIGNED HIE fixed-point search Claassen et al. (2013), p. 177	→ <code>discovery/dsep.py::hierarchy</code>	→ <code>test_fci_plus.py::test_hierarchy_recursively_adds_known_sepsets</code>
SOURCE-ALIGNED FCI+ Algorithm 2 Claassen et al. (2013), pp. 178-179	→ <code>discovery/fci_plus.py::FCIPlus.fit</code> → <code>discovery/dsep.py</code>	→ <code>test_published_reference_graphs.py</code>
SOURCE-ALIGNED Complete PAG orientation Zhang (2008), pp. 1873-1896	→ <code>discovery/rules.py::apply_orientation_rules[standard]</code>	→ <code>test_oracle_pag_rules.py</code>
ENGINEERING EXTENSION Finite-sample practical profile Repository design, inspected 2026-07-26	→ <code>config.py::FCIPlusConfig.practical</code>	→ <code>test_robust_application_profile.py</code>

Each row is backed by `reports/research/claim_evidence_matrix.csv`; the robust profile is deliberately separated from paper-aligned definitions.

Figure 5.1: Audit chain from the two primary sources to concrete repository symbols and executable validation. “Later completeness” is separated from the 2000 schedule, and practical robustness controls are labelled as engineering extensions.

5.11 Position relative to checked established software

The dated software audit distinguishes locally executed interfaces from documentation-only inspection. Local `fci_engine` FCI and FCI+, `pcalg` 2.7-12 FCI and `fciPlus`, and `causal-learn` 0.1.4.7 FCI completed small local smoke calls. Tetrad 7.6.10 was inspected in official source and release documentation but was not executed because a working Java runtime was unavailable.

The comparison supports an integration claim, not an accuracy ranking. `pcalg` exposes both FCI and FCI+, accepts user-defined CI tests, and its standard FCI result contains separator information. Its inspected `fciPlus` public result has materially less final separator provenance. `Causal-learn` exposes standard FCI and several named CI methods, but no public Claassen FCI+ entry point was identified in the inspected interface. Tetrad provides standard FCI and several distinct FCI-family and hybrid searches, an `IndependenceTest` interface, separator access, and broader GUI workflows; no drop-in Claassen FCI+ class was established.

The local package’s distinctive contribution is that paper-profile control, structured CI and orientation traces, separator-source provenance, D-SEP diagnostics, per-edge explanations, stability auditing, and JSON/CSV/HTML artifacts are available in one Python workflow. This is an auditability and integration improvement. It does not show that the local algorithm is universally faster or more accurate than those systems. Matched empirical comparisons must normalize endpoints semantically, hold data and CI settings as constant as possible, and separate standard FCI from FCI+ and from other FCI-family hybrids.

5.12 What implementation evidence does not establish

1. **Tests are not a general proof.** Passing unit tests and exact encoded fixtures establishes behavior on those cases, not correctness for every faithful MAG.

2. **A hand-authored target is not an independent oracle.** The Figure 4(b) MAG and exhibited separator are source based; the complete endpoint target is repository derived and requires independent equivalence-class validation before being called a published PAG.
3. **A paper profile is not an assumption checker.** It cannot verify faithfulness, causal acyclicity, exact CI information, or the true maximum MAG degree.
4. **Robustness is benchmark bounded.** Improved endpoint metrics or order stability on committed seeds and tested permutations do not imply universal finite-sample dominance.
5. **Auditability is not causal validity.** A complete trace can explain how a mark was produced while the underlying CI decision is still wrong.
6. **External agreement is not ground truth.** Agreement with pcalg or causal-learn is differential evidence about implementations, not proof of the causal graph.
7. **A PAG is not a treatment-effect analysis.** The output neither identifies one DAG nor estimates an adjustment set, intervention effect, mediation effect, or latent-variable value without additional assumptions and methods.

These limitations motivate the report’s separation between algorithm validation and the Tennessee STAR application. Exact graphical oracles test paper-aligned logic under known independence information; finite-sample benchmarks test empirical behavior under known simulated truth; established software supplies differential evidence; and the STAR experiment supplies a separate design-based basis for its strongest application inference.

Algorithm Validation

6.1 Exact oracle tests

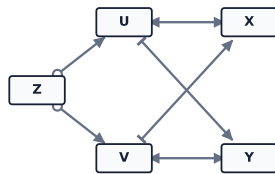
The strongest algorithm tests replace sample-based CI decisions with exact m-separation in a known MAG. This isolates graph-search correctness from sampling error. The regression suite contains three independently specified targets based on published structures:

1. the canonical Figure 4(b) D-SEP example from Claassen et al. (2013);
2. a latent-variable reference graph derived from the FCI literature (Spirtes et al., 2000);
3. an orientation-rule reference graph associated with completeness results in Zhang (2008).

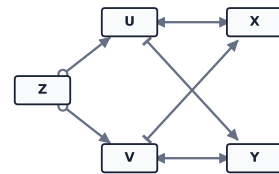
Both FCI and FCI⁺ recover the independently specified target PAGs on all three encoded fixtures. Figure 4(b) is additionally tested under four variable orders. The paper prints the Figure 4(b) MAG and exhibits $\{U, V, Z\}$ as a separator; it does not print the endpoint-complete target used by the repository.

Exact m-separation isolates algorithmic correctness from sampling error

**Repository-derived oracle PAG
for published Figure 4(b) MAG**



**Recovered FCI+ PAG
63 CI queries; exact endpoint match**



The X-Y candidate is removed by the hierarchical D-SEP stage using the separating set $\{U, V, Z\}$.

Figure 6.1: Exact-oracle validation on the central Figure 4(b) example. The source-grounded facts are the six MAG edges and the exhibited separator; the endpoint-complete comparison target is independently derived in the repository.

The initial adjacency stage retains the $X - Y$ link. FCI+ identifies it as a D-SEP candidate and removes it using $\{U, V, Z\}$. Standard FCI reaches the same final PAG through Possible-D-SEP. The measured query counts on this fixture are:

Table 6.1: Paper-aligned Figure 4(b) oracle comparison.

Algorithm	CI queries	Difference	Final PAG
Standard FCI	102	reference	Exact
FCI+	63	-39 (-38.2%)	Exact

These executed values are frozen in `reports/data/oracle_validation_summary.csv`. The artifact records the target PAG hash, `fa03c98d7851a4ba6327dcd136bc2076450451ff30cf312eeee6dbb2b71df7ea`, the exact-oracle profile, query counts, separator provenance, endpoint metrics, and execution status. It is generated programmatically from the repository’s MAG fixture; “completed” records successful execution, while `exact_target_recovered` separately records the scientific outcome.

6.2 Finite-sample integration test

Exact CI answers are necessary for testing the algorithm, but real data replace them with noisy statistical decisions. A seeded latent linear SEM based on the same reference graph provides the following integration result:

Table 6.2: Seeded finite-sample recovery on the Figure 4(b) SEM.

Sample size	FCI+	Standard FCI	Interpretation
$N = 5,000$	Extra $X \leftrightarrow Y$; exact F1 = 0.923	Exact PAG; exact F1 = 1.000	The sample CI decisions do not expose the FCI+ D-SEP candidate.
$N = 50,000$	Exact PAG; exact F1 = 1.000	Exact PAG; exact F1 = 1.000	Both algorithms recover this seeded fixture; this is not a universal sample threshold.

The same frozen artifact records these seeded runs with Fisher-Z, $\alpha = 0.001$, seed 1, conditioning and sparsity bounds of 3, and an unbounded path length. It includes all recovery metrics and logical CI-query counts: 102 and 52 at $N = 5,000$, and 102 and 63 at $N = 50,000$, for FCI and FCI+ respectively.

6.2.1 Multi-seed robustness regression

The application safeguard was also tested on 15 deterministic latent-SEM cases: five realistic graph families, three random seeds each, and 2,500 observations per case. The oracle PAG is known for

every case. This test compares the unchanged paper profile with the practical profile used in the STAR application.

Table 6.3: Mean recovery across 15 finite-sample oracle cases.

Profile	Skeleton F1	Exact F1	Semantic F1	Endpoint acc.	CI calls
Paper FCI+	0.979	0.536	0.951	0.744	197.1
Robust FCI+	0.985	0.730	0.985	0.840	401.9

The robust profile improves the mean skeleton and endpoint-oriented scores on these 15 runs. It uses about 2.04 times as many CI calls as paper FCI+. This does not prove universal dominance, but it provides a known-truth regression check that the STAR stability gain is not achieved by simply discarding useful structure.

This result is a useful warning against claiming that fewer CI tests imply uniformly better finite-sample accuracy. FCI and FCI⁺ can make different errors because they use different logical search routes after the initial skeleton.

6.3 Reference software

The STAR analysis runs CRAN `pcalg` 2.7-12 under R 4.5.3 as an independent reference implementation (Kalisch et al., 2012). The runner calls `pcalg::fciPlus` with selection-bias orientations enabled. It supplies a custom G-square test that exactly mirrors the Python compact-table rule: zero-marginal rows and columns are removed separately within each observed conditioning stratum, and the resulting likelihood-ratio statistics and degrees of freedom are summed. This avoids `disCITest`'s default minimum-cell shortcut, which would otherwise change the CI decisions rather than merely change the implementation.

The R output is treated as a differential reference, not ground truth. In particular, `pcalg::fciPlus` does not expose the local implementation's explicit $k = 3$ sparsity-bound parameter. Comparisons therefore report skeleton overlap, exact endpoint agreement, target-edge marks, runtime, CI calls, and cyclic-order sensitivity instead of awarding correctness by package name.

Key result. The oracle tests support the correctness of the implemented FCI and FCI⁺ search-and-orientation pipeline under exact conditional-independence information. The finite-sample tests simultaneously show that empirical results remain limited by CI-test error and tuning choices.

Comparison with Established Software

7.1 Two evidence layers

The comparison deliberately separates execution from interface inspection. The executed layer asks what happened on identical committed datasets under recorded settings. The capability layer asks what each public interface or official documentation exposes. Tetrad 7.6.10 (Tetrad Project, 2026) appears only in the latter as a *documentation-only comparison*: no empirical row or runtime is attributed to it because a working Java runtime was not available in the recorded environment.

All claims in this chapter are dated 26 July 2026. A named external package is a differential reference, not ground truth. Agreement can reveal translation or implementation errors; disagreement can reveal differences in CI semantics, search schedules, endpoint conventions, or finite-sample policies without identifying which output is causally true.

7.2 Executed known-truth benchmark

The benchmark contains five realistic latent-SEM scenarios, three deterministic repetitions per scenario, and 2,500 observations per run. Each method receives the same numeric matrix and variable order. The fixed paper/reference rows use Fisher-Z at $\alpha = 0.001$; robust FCI+ uses its documented sample-size rule, which resolves to $\alpha = 0.01$ here. The alpha mismatch is intentional because the row evaluates the public robust policy, not a literal Algorithm 2 reproduction.

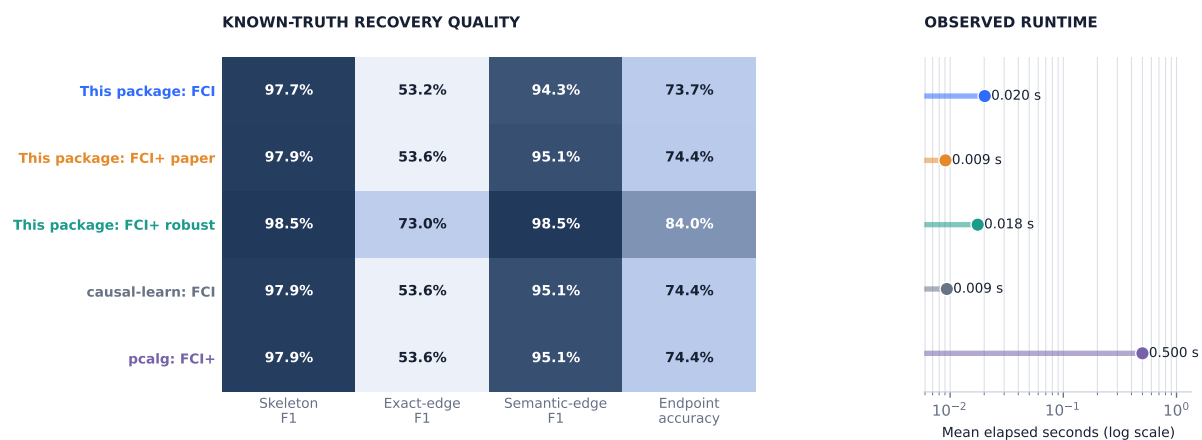
Table 7.1: Executed known-truth comparison: means over 15 completed runs.

Method (version)	Skeleton F1	Exact F1	Semantic F1	Endpoint acc.	Seconds	Logical CI
Local FCI paper (0.1.0)	0.977	0.532	0.943	0.737	0.020	449.7
Local FCI+ paper (0.1.0)	0.979	0.536	0.951	0.744	0.009	197.1
Local FCI+ robust (0.1.0)	0.985	0.730	0.985	0.840	0.018	401.9
causal-learn FCI (0.1.4.7)	0.979	0.536	0.951	0.744	0.009	not exposed
pcalg::fciPlus (2.7-12)	0.979	0.536	0.951	0.744	0.500	not exposed

Exact F1 requires both endpoint marks to match. Semantic F1 additionally credits endpoint patterns compatible with the target ancestral constraints, while endpoint accuracy scores individual marks. The

common conversion represents each PAG edge as an unordered node pair plus the two semantic endpoint marks; it never interprets raw package matrices as generic directed edges.

EXECUTED | 15 seeded known-truth cases per method



Descriptive, not a universal ranking: profiles and alpha policies differ by design. The pcalg timing includes R process startup; other timings are in-process calls.

Figure 7.1: Matched known-truth comparison. The robust profile improves the reported endpoint-oriented means on these scenarios while retaining a similar skeleton score and spending more logical CI queries than paper FCI+.

The equality of the aggregate local paper-FCI+, causal-learn, and pcalg scores is useful differential evidence on these fixtures. It is not an equivalence proof. Local and causal-learn times cover in-process algorithm calls, whereas the R timing includes process startup. `pcalg::fciPlus` and the local paper profile also do not expose identical controls for the true-degree bound k . Consequently, the table is not a language-neutral speed contest.

7.3 Capabilities and public audit surface

Table 7.2: Dated public-interface comparison. “Partial” means that some information can be obtained, but not through the same integrated result contract.

Tool	FCI	FCI+	Custom CI	Sepset provenance	Order/bootstrap	Structured audit/export
fci_engine 0.1.0	yes	yes	yes	yes, stage named	both	CI and orientation traces; D-SEP diagnostics; edge explanations; JSON/CSV/HTML
pcalg 2.7-12	yes	yes	yes	partial	order partial	standard-FCI sepsets; no equivalent bundled export established
causal-learn 0.1.4.7	yes	no public FCI+ found	named tests	not exported	not established	graph return; no equivalent structured trace established
Tetrad 7.6.10	yes	no Claassen FCI+ established	yes	yes	partial/broader workflow	logger and GUI facilities; not executed here

TOOL	Standard FCI	FCI+	Custom CI	Order audit	Bootstrap workflow	Orientation trace	Sepset provenance	Artifact export	EVIDENCE
fci_engine Python	●	●	●	●	●	●	●	●	EXECUTED
pcalg R	●	●	●	◐	?	?	◐	✗	EXECUTED
causal-learn Python	●	✗	✗	?	?	?	?	✗	EXECUTED
Tetrad Java	●	✗	●	◐	◐	◐	●	✗	DOC-ONLY

● Confirmed ◐ Partial / broader workflow ? Not established / not exported ✗ No

Figure 7.2: Executed status and checked capability surface. Tetrad is kept visually distinct because the comparison is documentation-only.

The checked interfaces support the bounded statement that the local package integrates paper-profile control, structured CI and orientation traces, separator-source provenance, D-SEP diagnostics, per-edge explanations, stability auditing, and artifact exports in one Python workflow. `pcalg` exposes both FCI and FCI+, accepts user-defined CI tests, and standard FCI exposes separator information. Causal-learn exposes standard FCI and named CI methods (causal-learn developers, 2026), but no inspected public Claassen FCI+ entry point. Tetrad provides FCI and several different FCI-family or hybrid algorithms; none was treated as a drop-in Claassen FCI+ implementation.

7.4 Fairness constraints and bounded improvement claim

Every comparison obeys five rules. First, circles, tails, and arrowheads are preserved in semantic normalization. Second, data, order, alpha, CI semantics, and search caps are held fixed where interfaces allow and every mismatch is disclosed. Third, standard FCI, Claassen FCI+, and other FCI-family hybrids are not relabelled as one algorithm. Fourth, logical queries, cache hits, executed statistical tests, and wall time remain different quantities. Fifth, smoke-call success establishes callability, not scientific validity.

Key result. The local package integrates traceability and robustness tooling that was not found as one public workflow in the checked interfaces. This is an auditability and integration improvement, not a universal accuracy or speed superiority claim.

The robust profile’s higher mean endpoint scores are therefore stated only “on this benchmark.” Its automatic alpha, stable deletion, max- p separator choice, conservative collider policy, and cautious tail orientation are engineering policies. They can trade endpoint specificity and query cost against tested stability; they are not revisions to the definition or theorem of Claassen et al. (2013).

Part II

Tennessee STAR Application

Study Design and Data

8.1 The Tennessee STAR experiment

The Tennessee Student/Teacher Achievement Ratio project was a large randomized experiment conducted in kindergarten through grade 3. Students and teachers were assigned to small classes, regular classes, or regular classes with a full-time aide. The official technical report describes small classes of approximately 13–17 students and regular classes of approximately 22–25 students (Word et al., 1990). Subsequent economic analysis has used STAR as a central source of evidence on class size and educational production (Krueger, 1999).

The present analysis uses the official Harvard Dataverse release (Achilles et al., 2008). The student-level file contains 11,601 records and 379 variables and is released under CC0 1.0. Data codes were checked against the accompanying 146-page user guide.

8.2 Why use causal discovery on a randomized experiment?

Randomization already provides a stronger basis for estimating the class-size effect than an unconstrained structure-learning algorithm. The purpose of FCI is therefore not to rediscover or replace randomization. It is to examine:

- whether grade-3 observation is structurally related to early achievement or background variables;
- whether early and later achievement remain adjacent after conditioning on the measured context;
- how the learned PAG changes when an early post-assignment outcome is included or excluded;
- whether FCI and FCI⁺ agree on skeletons and important adjacencies;
- how much search work FCI⁺ saves on a real dataset.

This design also provides an unusually useful external audit. If a data-only PAG implies an implausible pre-treatment direction or suggests latent confounding of a genuinely randomized assignment, the discrepancy should be reported rather than accepted literally.

Table 8.1: Design and observation layers kept separate in the STAR analysis.

Layer	Variables or operation	Inferential role
Assignment	Kindergarten class arm	Randomized design is the basis for the early arm contrast; the report does not reconstruct original blocks, switching, or compliance.
Pre-treatment context	Sex, ethnicity, age proxies, school context	Measured context for descriptive and discovery panels; not a claim that all common causes are observed.
Post-treatment outcomes	Kindergarten and grade-3 achievement	Kindergarten achievement may lie on a treatment pathway; including it changes the question answered by discovery.
Selection	Grade-3 observation indicator	Separate attrition panel examines whether later availability is structurally related to early variables.
Complete-case panels	Students with recorded grade-3 outcomes	Longitudinal and focused results condition on observation and are weaker than the early design-based contrast.

8.3 Cohort construction

The analysis begins with 6,325 students who have a kindergarten STAR class assignment, representing 79 kindergarten schools. Three panels are constructed for different questions.

Table 8.2: STAR analysis panels.

Panel	Rows	Nodes	Purpose
Attrition / observation	5,744	9	Relate kindergarten assignment, characteristics, context, and achievement to whether both grade-3 reading and mathematics scores are observed.
Longitudinal achievement	2,787	9	Study structure between kindergarten and grade-3 achievement among complete cases while retaining early achievement.
Focused treatment-outcome	2,976	8	Study the class-assignment/grade-3 relationship without allowing kindergarten achievement, an early post-assignment outcome, to serve as a separator.

The panels are intentionally not pooled. Each encodes a different statistical question, and including or excluding kindergarten achievement changes the meaning of separation between assignment and the later outcome.

8.4 Variable coding

Table 8.3: Variables supplied to FCI and FCI+.

Analysis variable	Source fields	Coding
K_Class	gkclasstype	Small, regular, or regular with aide.
Gender	gender	Male or female, using the official data codes.
Ethnicity	race	White, Black, or Other; smaller source categories are collapsed to avoid extremely sparse contingency cells.
Free_Lunch	gkfreelunch	Free/reduced-price lunch versus non-free lunch.
School_Context	gksurban	Inner city, suburban, rural, or urban.
Entry_Age	birthyear, birthmonth	Age in months at September 1985, converted to four quantile categories.
Teacher_Experience	gktyears	0–5, 6–10, 11–20, or 21+ years.
K_Achievement	gktreadss, gktmathss	Reading plus mathematics scaled score, converted to kindergarten achievement quartiles.
Grade3_Achievement	g3treadss, g3tmathss	Reading plus mathematics scaled score, converted to grade-3 achievement quartiles among students with both scores.
Grade3_Observed	g3treadss, g3tmathss	Indicator that both grade-3 scores are present.

All model inputs are integer-coded discrete categories. The exact numeric tables supplied to the algorithms are committed under `case_studies/tennessee_star/data/processed`.

STAR Analysis Methods

9.1 Descriptive randomized-arm reference

Before causal discovery, the report computes observed means and randomized-arm contrasts for:

1. the sum of kindergarten reading and mathematics scaled scores;
2. the sum of grade-3 reading and mathematics scaled scores among students with both scores;
3. the probability that both grade-3 scores are observed.

Uncertainty is estimated with 1,000 bootstrap samples of kindergarten schools. Resampling schools rather than individual students preserves the main within-school dependence in the descriptive comparison. These contrasts are reported separately from FCI output because they use the experiment’s assignment as the basis for interpretation.

9.2 Discrete G-square conditional-independence test

The application uses the likelihood-ratio G-square test. For variables X and Y , conditioning set S , observed contingency counts O , and expected counts E under $X \perp Y \mid S$, the statistic is

$$G^2 = 2 \sum_s \sum_{x,y} O_{xys} \log \left(\frac{O_{xys}}{E_{xys}} \right),$$

with zero-count contributions omitted. The test is appropriate for the discretised panels, but high-order conditioning can create sparse cells. The primary FCI⁺ analysis uses $k = 3$, while the sensitivity grid repeats the focused analysis at $k \in \{2, 3, 4\}$. These are bounded analysis settings, not evidence that the unknown STAR MAG has a known maximum degree.

9.3 Primary discovery configuration

Table 9.1: Primary STAR discovery configuration.

Setting	Value
CI test	G-square on discretised variables
Significance threshold	$\alpha = 0.05$
Standard FCI	<code>profile="paper"</code> ; unbounded paper schedule
Paper FCI ⁺	<code>profile="paper"</code> ; $k = 3$
Robust application FCI ⁺	<code>profile="practical"</code> ; stable skeleton; strongest separating set; conservative colliders; robust orientation; depth and sparsity bound 3
R reference	<code>pcalg::fciPlus</code> ; selection bias enabled; no exposed explicit k setting
Timing	Three full-data repetitions; median elapsed time reported
PAG stability	12 kindergarten-school cluster bootstrap fits per algorithm and panel for all three self-implemented profiles
Order audit	Every cyclic column shift for all local profiles and the R reference
Random seed	Fixed by the application script for reproducibility
Prior temporal constraints	None supplied to discovery; temporal reversals are audited after fitting

Not imposing temporal or randomization constraints allows the analysis to test the behaviour of the data-only algorithm. It also means that implausible orientations must not be treated as credible causal chronology.

9.4 School-cluster PAG stability

For each panel and algorithm, the 79 kindergarten schools are sampled with replacement. All student rows from a selected school are included, including repeated copies when a school is sampled more than once. FCI or FCI⁺ is then refitted and the presence of each unordered adjacency is recorded.

With $B = 100$ replicates, the estimated adjacency frequency is

$$\hat{\pi}_{XY} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{X \text{ and } Y \text{ are adjacent in replicate } b\}.$$

This is an exploratory stability diagnostic with one-percentage-point frequency resolution. It is not a confidence interval, a posterior edge probability, or an endpoint-compatible uncertainty summary.

9.5 Sensitivity analysis

The focused treatment-outcome analysis is repeated for:

- $\alpha \in \{0.01, 0.05\}$;
- three versus four quantile bins for age and achievement;

- standard FCI, paper-profile FCI⁺, and robust application FCI⁺.

The target is whether `K_Class` remains adjacent to `Grade3_Achievement`. This directly tests whether the most substantively tempting PAG edge is robust to reasonable analysis choices.

9.6 Decision rule for robust conclusions

The application distinguishes a reproducible adjacency from a credible direction. A skeleton-level conclusion is promoted only when the target adjacency survives every cyclic column shift and has high school-cluster bootstrap frequency. A directional conclusion additionally requires invariant endpoint marks, no conflict with measurement time or randomization, and no material reversal under the alpha/binning sensitivity checks.

Failure of the directional rule does not delete the edge. It downgrades the claim to conditional-dependence structure with unresolved direction. This pre-specified rule is why the robust profile's circle endpoints are treated as a reliability improvement rather than an incomplete result.

STAR Results

10.1 Observed randomized-arm outcomes

Table 10.1: Observed outcome summaries by kindergarten class arm. Intervals use 1,000 kindergarten-school bootstrap samples.

Outcome	Arm	<i>n</i>	Estimate	95% interval
Kindergarten score	Regular	2,005	918.04	908.78 to 927.05
	Regular + aide	2,043	918.36	908.52 to 927.07
	Small	1,738	931.94	922.70 to 940.12
Grade-3 score	Regular	1,047	1246.74	1238.96 to 1254.01
	Regular + aide	1,019	1247.25	1238.53 to 1255.48
	Small	936	1258.43	1250.58 to 1266.17
Grade-3 observed	Regular	2,194	47.72%	42.72% to 52.14%
	Regular + aide	2,231	45.67%	40.64% to 50.39%
	Small	1,900	49.26%	44.46% to 53.77%

Grade-3 score summaries condition on both reading and mathematics scores being observed. Kindergarten score summaries also exclude missing score components, so they are design-supported observed-outcome comparisons rather than a fully reconstructed intention-to-treat estimand.

Randomized-arm descriptive contrasts with 95% school-cluster intervals

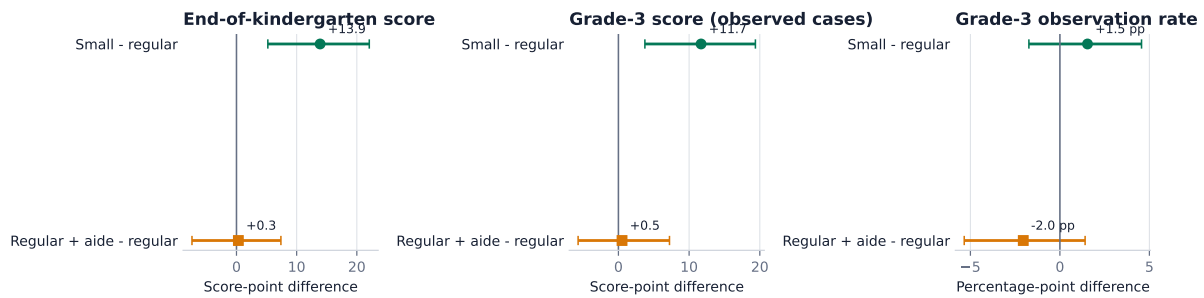


Figure 10.1: Contrasts against the regular-class arm. Small classes show positive score differences at kindergarten and grade 3 among observed cases; the aide arm remains close to zero.

The small-minus-regular kindergarten contrast is 13.90 points, with a 95% school-cluster interval from 5.21 to 22.07. The regular-with-aide contrast is 0.31 points, with an interval from -7.40 to 7.38 .

At grade 3 among observed students, the small-class contrast remains positive at 11.69 points (3.74 to 19.38). The aide contrast is 0.51 points (-5.70 to 7.22). Observation rates themselves do not show a precise class-arm difference: small minus regular is $+1.54$ percentage points (-1.73 to 4.56), while aide minus regular is -2.05 points (-5.35 to 1.41).

Key result. The observed randomized-arm evidence supports a small-class achievement advantage and does not support a comparable advantage from adding an aide to a regular class. The grade-3 score contrast is conditional on later observation and therefore has a weaker causal interpretation than the early outcome comparison.

10.2 Attrition and grade-3 observation

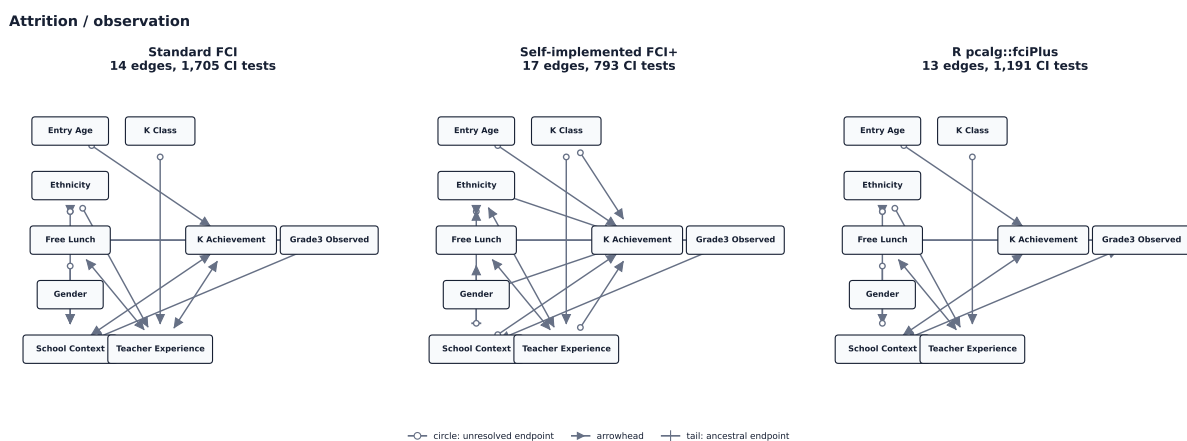


Figure 10.2: Three PAGs for the attrition/observation panel: self-implemented FCI, self-implemented FCI⁺, and R `pcalg::fciPlus`. The skeletons substantially overlap, while the achievement/observation endpoint differs.

Standard FCI returns 14 edges after 1,705 CI tests; self-implemented FCI⁺ returns 17 edges after 793 tests; and R `pcalg::fciPlus` returns 13 edges after 1,191 tests. The FCI/self-FCI⁺, FCI/R-FCI⁺, and

self-/R-FCI⁺ skeleton Jaccard values are 0.824, 0.929, and 0.765. All three omit a direct `K_Class`–`Grade3_Observed` adjacency. In the self-implementation school bootstrap that adjacency appears in 12% of FCI fits, 16% of paper FCI⁺ fits, and 15% of robust FCI⁺ fits.

By contrast, `K_Achievement`–`Grade3_Observed` is adjacent in 100% of the bootstrap fits for all three local profiles. The corresponding free-lunch/observation adjacency is also stable in the targeted audit. These results suggest that later data availability is not independent of early achievement and measured socioeconomic or school context.

The three endpoint readings are revealing. Standard FCI gives the impossible *grade-3 observation* → *kindergarten achievement*; self-FCI⁺ leaves a circle at the later endpoint; and R-FCI⁺ gives the temporally admissible *kindergarten achievement* → *grade-3 observation*. The R result is more plausible on this edge, but package identity alone cannot establish that its orientation is statistically correct.

Interpretation caution. The attrition PAG does not prove a particular missingness mechanism. It shows that the observed conditional-independence structure links grade-3 availability to earlier achievement and context. A formal missing-data analysis would require an explicit estimand and additional assumptions.

10.3 Longitudinal achievement

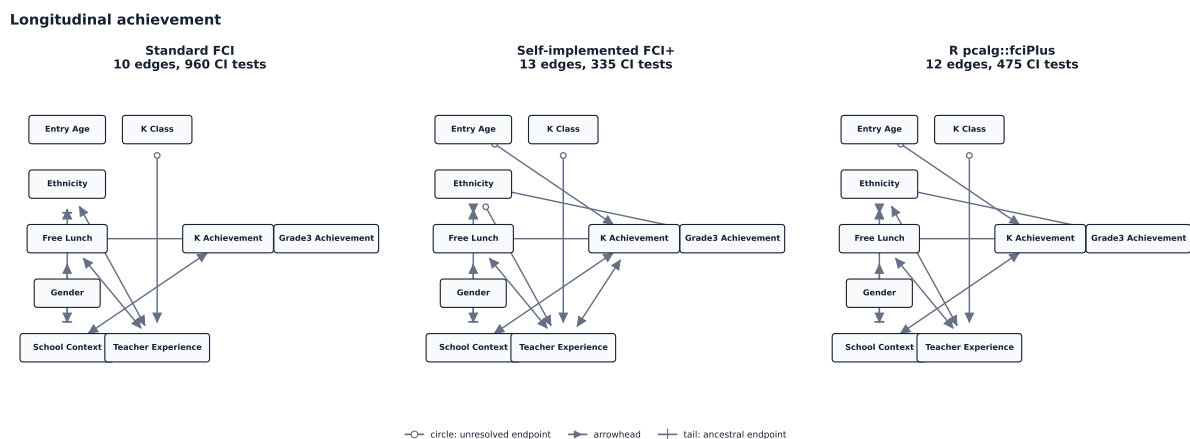


Figure 10.3: Three PAGs for the longitudinal complete-case panel. All retain the kindergarten/grade-3 achievement adjacency, but the two FCI⁺ implementations orient it against chronology.

Standard FCI returns 10 edges and 960 CI tests. Self-FCI⁺ returns 13 edges and 335 tests, while R-FCI⁺ returns 12 edges and 475 tests. Pairwise skeleton Jaccard is 0.769 for FCI/self-FCI⁺, 0.833 for FCI/R, and 0.923 for the two FCI⁺ implementations.

All three algorithms retain the kindergarten/grade-3 achievement adjacency, and it appears in all 100 school-cluster bootstrap samples for all three local profiles. Neither full-data PAG retains a direct class-assignment/grade-3 achievement adjacency once kindergarten achievement is included. That treatment-outcome adjacency nevertheless reappears in 71% of FCI, 79% of paper FCI⁺, and 80% of robust-profile bootstrap fits despite being absent from each corresponding full-data PAG. This discordance is a warning that conservative orientation and order invariance do not remove school-resampling sensitivity.

Both FCI⁺ implementations orient grade-3 achievement toward kindergarten achievement, which is

temporally impossible if read causally. Standard FCI leaves the grade-3 endpoint unresolved. The agreement between two implementations is therefore not sufficient to override chronology: it may replicate the same finite-sample CI pattern and orientation logic under complete-case selection.

10.4 Focused treatment-outcome panel

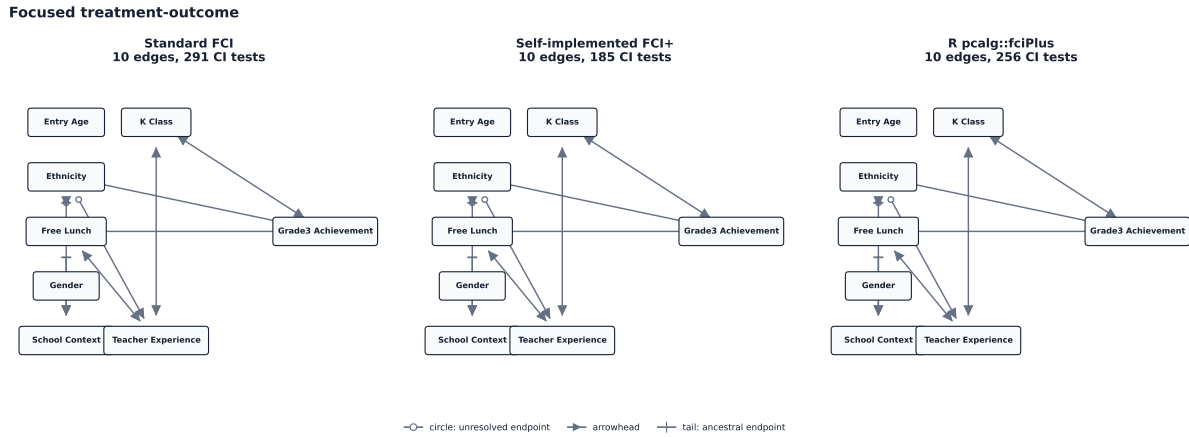


Figure 10.4: PAGs for the focused complete-case panel. All three algorithms recover the exact same 10-edge skeleton and endpoint pattern, including the bidirected class-assignment/grade-3 edge.

Removing kindergarten achievement from the node set changes the result. Standard FCI, self-FCI⁺, and R-FCI⁺ all return the same 10-edge PAG, not merely the same skeleton. All three contain

$$K_Class \leftrightarrow Grade3_Achievement.$$

The adjacency appears in 92% of the FCI school bootstraps, 85% of the paper self-FCI⁺ bootstraps, and 96% of the robust-profile bootstraps. The independent exact PAG replication makes this observed conditional-independence pattern more credible, but does not make the literal latent-confounding interpretation credible.

A naive reading would call the bidirected edge evidence of latent confounding. That reading is not defensible here. Kindergarten class assignment was randomized, while the focused panel conditions on grade-3 outcomes being observed and omits early post-assignment achievement. Conditioning on observation, school-level dependence, omitted post-randomization variables, and CI-test approximation can all create an observed independence pattern that is represented by arrowheads at both endpoints.

10.5 Robust application profile

Robust FCI+ application profile: conservative STAR PAGs

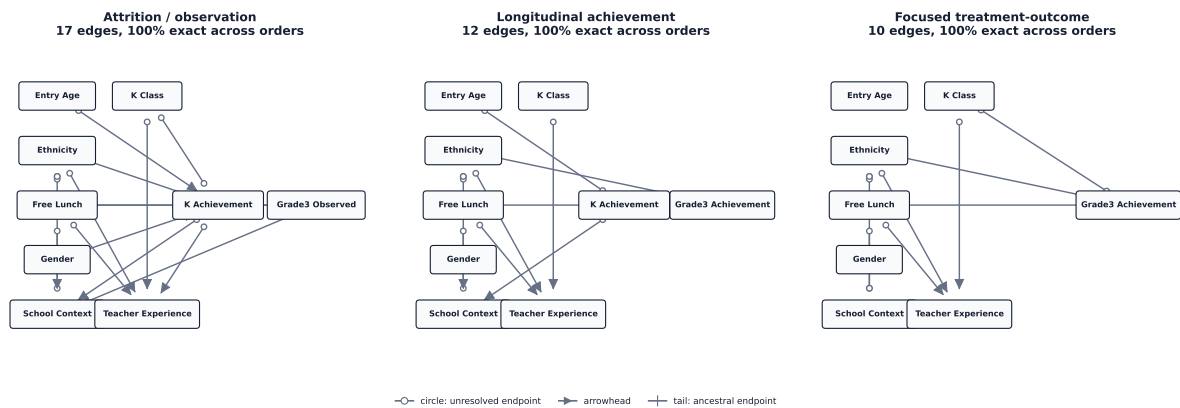


Figure 10.5: The three STAR PAGs from the separate robust application profile. Stable skeleton search and conservative orientation preserve the main adjacencies while leaving unsupported directions unresolved.

The robust profile is not presented as a fourth paper implementation. It is a finite-sample decision layer applied after the literal FCI and FCI+ comparisons. It returns 17, 12, and 10 edges on the attrition, longitudinal, and focused panels, respectively. The target relations are

$$\begin{aligned} K_Achievement &\circ\circ Grade3_Observed, \\ K_Achievement &\circ\circ Grade3_Achievement, \\ K_Class &\circ\circ Grade3_Achievement. \end{aligned}$$

Thus it retains the relevant skeleton information without repeating the paper-profile chronology reversal or the focused-panel bidirected over-interpretation.

Table 10.2: Cyclic-order audit for paper and robust self-FCI+ profiles.

Panel	Profile	Exact PAG	Worst skeleton	Temporal reversals
Attrition	Paper	100.0%	100.0%	0
	Robust	100.0%	100.0%	0
Longitudinal	Paper	55.6%	92.3%	1
	Robust	100.0%	100.0%	0
Focused	Paper	75.0%	100.0%	0
	Robust	100.0%	100.0%	0

Key result. The robust profile improves the measured reliability of the STAR output: it returns the same full PAG object for every tested cyclic order, every target adjacency survives, and no fully directed edge runs backward in measurement time. The price is deliberate under-orientation and additional CI work. Reliability has improved; identifiability has not.

10.6 Stability and sensitivity

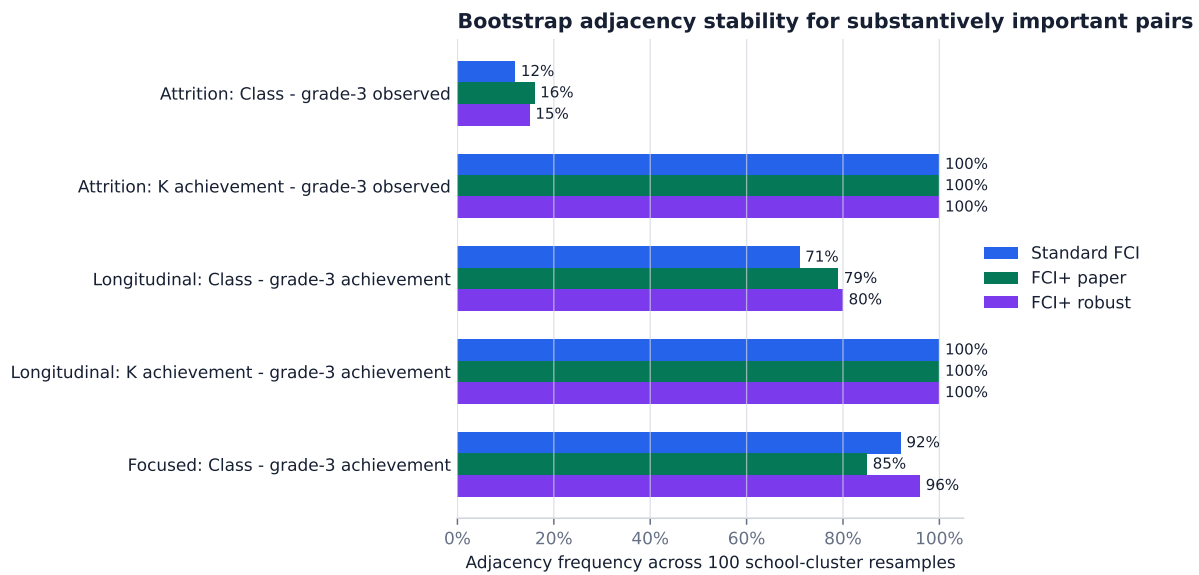


Figure 10.6: School-cluster adjacency frequencies for selected pairs. Frequencies measure skeleton stability only; they do not validate endpoint orientation.

Sensitivity of the focused class-assignment / grade-3 adjacency

Standard FCI	Absent 225 CI tests	Present 305 CI tests	Absent 252 CI tests	Present 291 CI tests
FCI+ paper, k=2	Absent 146 CI tests	Present 203 CI tests	Absent 152 CI tests	Present 174 CI tests
FCI+ paper, k=3	Absent 150 CI tests	Present 214 CI tests	Absent 163 CI tests	Present 185 CI tests
FCI+ paper, k=4	Absent 150 CI tests	Present 214 CI tests	Absent 163 CI tests	Present 185 CI tests
FCI+ robust, k=2	Absent 217 CI tests	Present 316 CI tests	Absent 223 CI tests	Present 290 CI tests
FCI+ robust, k=3	Absent 233 CI tests	Present 360 CI tests	Absent 253 CI tests	Present 334 CI tests
FCI+ robust, k=4	Absent 234 CI tests	Present 367 CI tests	Absent 257 CI tests	Present 341 CI tests
	3 bins $\alpha = 0.01$	3 bins $\alpha = 0.05$	4 bins $\alpha = 0.01$	4 bins $\alpha = 0.05$

Figure 10.7: Focused treatment-outcome sensitivity. The adjacency appears at $\alpha = 0.05$ and disappears at $\alpha = 0.01$, regardless of whether three or four quantile bins or $k = 2, 3, 4$ is used for FCI+.

The sensitivity result is decisive for interpretation. The focused treatment-outcome adjacency is robust to the tested bin count and FCI+ degree bound but not to the CI threshold. At $\alpha = 0.01$, none of the local configurations retains it; at $\alpha = 0.05$, the paper profiles retain a bidirected mark for every tested k . The robust endpoint is $K_Class \leftarrow \circ Grade3_Achievement$ with three bins and $K_Class \circ \circ Grade3_Achievement$ with four bins.

The stronger threshold requires more evidence against conditional independence before an edge is kept. Its removal therefore indicates that the edge is driven by relatively weak conditional-dependence evidence,

not an invariant feature across reasonable test settings.

10.7 Cross-implementation agreement and order audit

Cross-implementation agreement is stronger for skeletons than endpoints

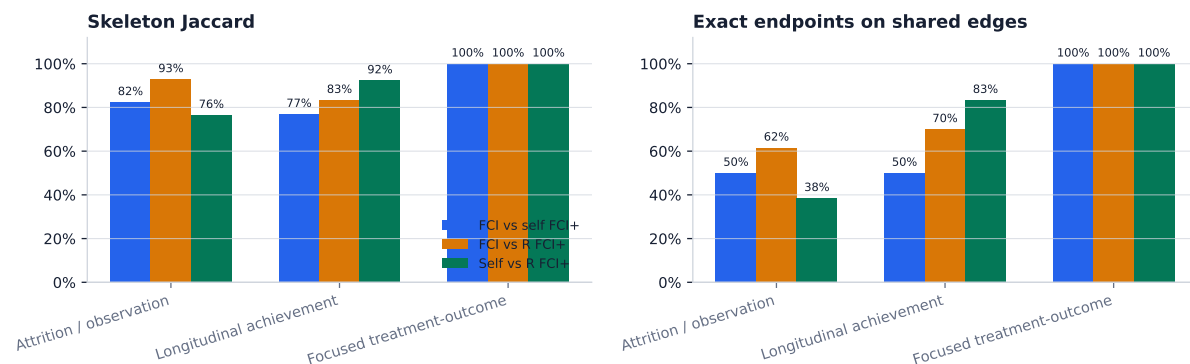


Figure 10.8: Pairwise skeleton Jaccard and exact endpoint agreement on shared edges. Skeleton agreement is consistently stronger than endpoint agreement outside the focused panel.

Table 10.3: Three-algorithm structural agreement. Endpoint rates use only adjacencies shared by the indicated pair.

Panel	Pair	Skeleton Jaccard	Exact endpoint rate
Attrition	FCI / self-FCI ⁺	0.824	50.0%
	FCI / R-FCI ⁺	0.929	61.5%
	self- / R-FCI ⁺	0.765	38.5%
Longitudinal	FCI / self-FCI ⁺	0.769	50.0%
	FCI / R-FCI ⁺	0.833	70.0%
	self- / R-FCI ⁺	0.923	83.3%
Focused	FCI / self-FCI ⁺	1.000	100.0%
	FCI / R-FCI ⁺	1.000	100.0%
	self- / R-FCI ⁺	1.000	100.0%

Cyclic variable-order robustness separates skeletons from endpoints

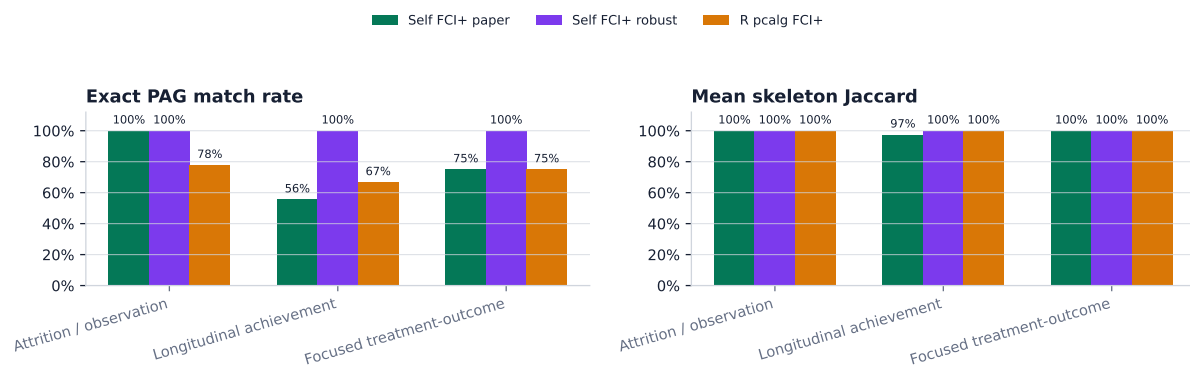


Figure 10.9: Cyclic-order audit for paper self-FCI⁺, robust self-FCI⁺, and R pcalg::fciPlus. Skeleton agreement is generally stronger than exact endpoint agreement; the robust profile is invariant on both measures.

The order audit rotates the input variable list through every cyclic shift. The R reference preserves every panel skeleton, while reproducing exact PAG endpoints in 7 of 9 attrition orders, 6 of 9 longitudinal orders, and 6 of 8 focused orders. Paper self-FCI⁺ is exact in 9 of 9, 5 of 9, and 6 of 8 orders, respectively; its longitudinal skeleton changes slightly in some orders. Robust self-FCI⁺ is exact in all 26 refits. The combined evidence supports skeleton robustness while independently showing that more aggressive endpoint marks can be implementation- and order-sensitive.

10.8 Computational comparison

STAR fits: paper validation and robust application profile

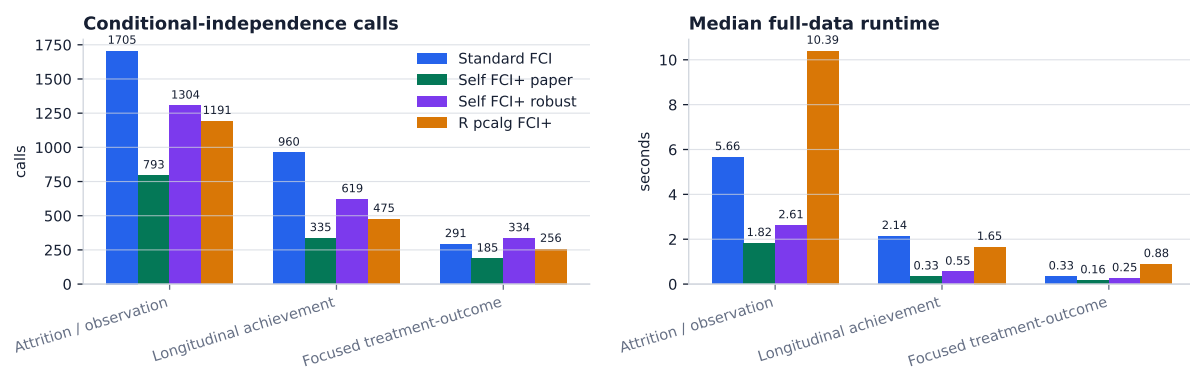


Figure 10.10: Full-data search work and median runtime. All algorithms use the same panel, compact-table G-square test, and $\alpha = 0.05$; internal search schedules and language runtimes are not identical.

Table 10.4: Paper-comparison and robust-profile performance on the STAR panels.

Panel	Algorithm	Edges	CI tests	Median seconds
Attrition	FCI	14	1,705	5.437
	Self-FCI ⁺ paper	17	793	1.788
	Self-FCI ⁺ robust	17	1,304	2.838
	R-FCI ⁺	13	1,191	10.394
Longitudinal	FCI	10	960	2.148
	Self-FCI ⁺ paper	13	335	0.337
	Self-FCI ⁺ robust	12	619	0.562
	R-FCI ⁺	12	475	1.650
Focused	FCI	10	291	0.322
	Self-FCI ⁺ paper	10	185	0.159
	Self-FCI ⁺ robust	10	334	0.247
	R-FCI ⁺	10	256	0.884

FCI⁺ uses 46.5%, 34.9%, and 63.6% of the standard FCI CI-test counts on the attrition, longitudinal, and focused panels. The independent R FCI⁺ uses 69.9%, 49.5%, and 88.0% of the corresponding standard-FCI counts. Runtime is not a language-neutral benchmark: the R runner includes R-level contingency-table construction, whereas the Python implementation uses the package's native data path. CI-call counts are the more interpretable search-work comparison.

Relative to paper self-FCI⁺, the robust profile uses 1.64, 1.85, and 1.81 times as many CI calls. Relative to standard FCI, it uses 76.5%, 64.5%, and 114.8%. The added work purchases stable skeleton deletion and invariant conservative endpoints; it is not claimed as a speed improvement.

These results demonstrate reduced search work, not universal algorithmic dominance. In the two panels where skeleton Jaccard is below one, efficiency and empirical structure differ simultaneously.

Causal Interpretation and Evidence Hierarchy

11.1 Five tiers of evidence

The application uses an explicit hierarchy so that a visually definite PAG mark cannot outrank the experimental design.

Table 11.1: Evidence hierarchy used for every STAR claim.

Tier	Evidence	Permitted claim
1	Randomized assignment and school-cluster arm contrast	Bounded causal statement for the analyzed observed outcome, qualified by missingness, compliance, and unreconstructed assignment blocks.
2	Adjacency replicated across methods, resamples, tested orders, and sensitivity settings	Robust skeleton-level structural association or constraint; no magnitude or direction claim.
3	Stable skeleton with circles or endpoint disagreement	Adjacency is more credible than orientation; endpoint remains unidentified.
4	Implementation-, order-, or chronology-sensitive endpoint	Diagnostic evidence only; not a substantive direction.
5	Threshold-sensitive or complete-case exploratory pattern	Hypothesis for external validation, not a causal conclusion.

FCI returns equivalence-class constraints and does not estimate a treatment effect. The randomized contrasts and the learned PAGs therefore answer different questions even when they concern the same variables.

11.2 What can be concluded

11.2.1 Design-supported early observed-arm contrast

STAR's random assignment is the primary causal identification mechanism. Within the kindergarten cohort and among students with observed score components, the small-class arm is 13.90 points above the regular arm, with a school-cluster 95% interval of 5.21 to 22.07. This design-supported observed-arm contrast is consistent with a beneficial early effect. It is not an identified causal effect in the post-assignment observed-outcome subset without additional assumptions about missing outcomes, compliance, and the original randomization blocks.

The PAG is consistent with an assignment/achievement connection in the FCI+ attrition panel, but this bounded interpretation does not depend on that endpoint. It depends on the experimental design and the stated selection qualification.

11.2.2 Adding an aide does not show the same advantage

The regular-with-aide contrasts are close to zero at kindergarten and grade 3, and both intervals include zero. In these observed data there is no evidence that adding an aide to a regular-size class reproduces the achievement advantage of reducing class size.

11.2.3 Early achievement is consistently retained at the skeleton level

Kindergarten and grade-3 achievement remain adjacent in the full longitudinal fits of all three algorithms and in 100% of the self-implementation school bootstraps. Kindergarten achievement is also adjacent to grade-3 observation in all three attrition fits. Cross-implementation agreement therefore supports a robust *skeleton-level* relationship between early performance, later performance, and later data availability.

This finding still does not quantify mediation. Kindergarten achievement is a post-assignment variable, and a formal mediation analysis would require additional counterfactual assumptions. Moreover, the direction of the longitudinal achievement edge fails the chronological audit for both FCI+ implementations.

11.2.4 Observation has stable structural links beyond class arm

The full-data attrition PAGs do not retain a direct class-arm/observation adjacency, and its bootstrap frequency is low. Observation is more stably connected to early achievement and measured context. The evidence therefore supports concern about selective attrition without showing a stable direct effect of class arm on grade-3 observation.

11.3 What cannot be concluded

1. **No unique DAG is identified.** Every PAG represents an equivalence class.
2. **No causal effect magnitude comes from FCI.** The score differences are arm contrasts; PAG edges contain no “points gained” coefficient.
3. **A bidirected edge is not proof of a hidden confounder.** In the focused panel it conflicts with the known randomization unless selection, post-treatment conditioning, or other violations are considered.

4. **The grade-3 contrast is not a complete long-run intention-to-treat estimate.** It conditions on observed grade-3 outcomes.
5. **Temporal reversals are not credible chronology.** They are algorithm diagnostics under violated or approximate assumptions.
6. **Implementation agreement is not proof.** Shared CI decisions and shared logical rules can reproduce the same wrong endpoint.
7. **Tested cyclic shifts are not all permutations.** Exact reproduction over 26 recorded cyclic orders is a finite audit, not a proof of general order invariance.
8. **Bootstrap frequency is not a posterior probability.** It describes adjacency recurrence under school resampling and the fixed analysis pipeline.
9. **Absence is policy conditional.** Failure to retain an edge under one node set, CI test, and threshold is not proof of zero causal effect.

Key result. The strongest application result is a design-supported observed-arm contrast: smaller kindergarten classes have higher observed early achievement than regular classes, while the aide arm shows no comparable contrast. The positive interval is consistent with a beneficial early effect, but it does not by itself identify a causal effect in the selected observed-outcome subset. The main contribution of the three-algorithm comparison is to separate replicated skeleton structure from unstable endpoint orientation. It exposes selection, persistence, and specification sensitivity around the experiment; it does not replace it.

Critical Self-Assessment

12.1 Strengths

- The algorithm and application are physically and logically separated.
- The data source, user guide, exact model inputs, random seeds, result tables, and report artifacts are committed.
- Standard FCI is always reported alongside FCI⁺.
- An independently maintained R implementation is run with matched CI semantics and its disagreements are retained in the report.
- Randomized-arm evidence is separated from exploratory PAG evidence.
- School-level resampling acknowledges the most important clustering unit.
- Endpoint reversals and threshold sensitivity are reported as failures of literal interpretation rather than silently corrected.
- The software is validated with exact oracle graphs, finite-sample integration tests, strict typing, multi-version tests, and wheel installation checks.

12.2 Statistical limitations

12.2.1 Discretisation and sparse contingency tables

Achievement and age are reduced to quantile categories, and teacher experience is binned. This enables G-square testing across mixed educational variables but discards within-bin information. As conditioning sets grow, some multiway contingency cells may have small expected counts, weakening the chi-square approximation.

12.2.2 Dependence within schools and classrooms

The primary CI test treats student rows as independent. The bootstrap resamples schools, which measures sensitivity to school composition, but it does not turn each underlying G-square test into a cluster-robust CI test. Classroom identifiers and multilevel structure should be incorporated in a more specialised analysis.

12.2.3 Complete-case selection

The longitudinal and focused panels require grade-3 scores. If observation is affected by prior achievement, mobility, family circumstances, or school processes, conditioning on complete cases may open selection paths. The attrition panel provides evidence that observation is structurally related to early variables, reinforcing this concern.

12.2.4 Post-treatment variables

Kindergarten achievement and teacher experience are measured during the treatment period. They are not baseline covariates in the same sense as sex or ethnicity. Including kindergarten achievement may block part of the class-size pathway; excluding it may leave omitted post-treatment dependence. The two panels deliberately show both specifications instead of selecting the one with the preferred edge.

12.2.5 Faithfulness and finite-sample CI error

FCI-family guarantees rely on faithfulness and correct CI decisions. Near-zero or cancelling effects may violate faithfulness, while finite samples produce false dependence and false independence. The threshold-sensitive focused edge is a direct example of this limitation.

12.2.6 Unknown sparsity bound

The FCI⁺ paper profile uses $k = 3$. The true maximum degree of the underlying STAR MAG is not observed and cannot be verified from the learned graph alone. If the required separating set exceeds the bound, FCI⁺ may retain an extra edge. Conversely, `pcalg::fciPlus` does not expose a directly comparable k argument, so the independent reference is not an identical internal search schedule.

12.2.7 Endpoint order sensitivity

The R and paper-profile self-FCI⁺ audits change some endpoint marks; the paper self-FCI⁺ longitudinal skeleton also changes slightly. This is a practical warning for constraint-based orientation under finite-sample CI decisions: deterministic output for one column order is not the same as order-invariant causal information. The robust profile removes the observed order dependence by stable deletion and conservative orientation. The report therefore uses it for applied interpretation, while still giving more evidential weight to skeleton consensus than to isolated endpoint marks.

12.2.8 PAG adjacency bootstrap

One hundred school-cluster graph bootstrap replicates improve the earlier 12-run pilot to one-percentage-point frequency resolution. They remain adjacency-stability diagnostics rather than confidence intervals or posterior probabilities that an edge is true; endpoint-compatible uncertainty and Monte Carlo error are not estimated.

12.2.9 Software maturity and validation coverage

The package is version 0.1.0 and classified pre-alpha. Strict typing covers the project’s own interfaces, but third-party numerical and graph libraries do not all provide equally complete type information. Unit tests and hand-authored oracle targets are strong regression evidence, not an independent formal proof or an exhaustive MAG-equivalence checker. In particular, the complete Figure 4(b) PAG target is repository derived even though its MAG and exhibited separator are source grounded.

12.2.10 External-software comparability

The known-truth comparison normalizes endpoint semantics and controls inputs, but the interfaces do not expose identical search caps, query counters, or timing boundaries. The robust row also intentionally uses a different alpha policy. External agreement is therefore differential evidence, while cross-language wall time is descriptive only. Tetrad capability statements were documentation checked and not executed.

12.3 Exact-oracle node-count scaling audit

An additional deterministic audit runs the two local paper profiles on a directed chain and on the Figure 4(b) D-SEP core augmented with isolated observed variables, at 5, 10, 20, 40, and 80 nodes. Every recorded run recovers the exact target skeleton. On the five-node D-SEP core, self-FCI⁺ uses 62 CI queries versus 101 for standard FCI. That advantage does not persist after irrelevant isolated variables are added: at 80 nodes the recorded counts are 5,162 versus 3,251. On the 80-node chain, which has no difficult D-SEP link, the counts are 24,338 versus 12,638.

This is not a contradiction of the fixed- k polynomial theorem. The theorem is a worst-case upper bound, not a pairwise dominance result, and augmentation has a real cost when targeted D-SEP search offers little benefit. The scaling audit therefore sharpens the performance claim: paper FCI+ reduces queries on the canonical hard link and all three primary STAR panels, but it is not uniformly faster. Full rows and timing metadata are stored in `reports/data/scaling_benchmark.csv`.

12.4 Substantive limitations

Limitation. The STAR application demonstrates an auditable causal-structure workflow; it is not a definitive re-estimation of the experimental treatment effect, and known-impossible orientations are reported as diagnostics.

Original randomization blocks, class switching, compliance, and school-specific assignment procedures are not reconstructed; a policy estimate must return to those records and define an intention-to-treat estimand. The transparent composite score simply adds reading and mathematics, leaving test scaling, subject-specific, distributional, and subgroup effects unexamined.

Conclusions and Future Work

13.1 Conclusions

The project achieved its two-stage objective and answers the five research questions with three bounded findings.

First, source fidelity is explicit and testable. Standard FCI maps to the historical 2000 search schedule, whose Theorem 6.4 is a correctness result for a partially oriented inducing-path graph under correct CI information, not a modern maximal-orientation theorem. FCI+ maps to Claassen et al.’s augmented skeleton and hierarchy in Algorithm 2; its polynomial result remains the restricted $O(N^{2(k+2)})$ exact-oracle query bound for fixed true degree k , faithfulness, and constant-time CI. The source-to-code-to-test matrix makes literal paper mechanisms, later orientation completeness, neutral software structure, and robust engineering policy visibly different.

Second, validation is layered rather than circular. Exact m-separation tests recover the independently specified targets on the encoded fixtures, including hierarchical removal of the Figure 4(b) false link with the source-exhibited separator $\{U, V, Z\}$. The known-truth benchmark executes five methods over 15 runs: robust FCI+ improves the recorded mean endpoint metrics, while paper FCI+ reduces logical queries relative to standard FCI. The causal-learn and pcalg comparisons supply differential evidence, and Tetrad remains documentation-only. The package’s defensible improvement is its integrated audit surface, not a claim to be universally fastest or most accurate.

Third, the package was applied to a real educational experiment without mixing domain code into the algorithm. The STAR results show a clear observed small-class achievement advantage, no comparable aide advantage, stable early-to-later achievement structure, and evidence that later observation is selective. An independent `R pcalg::fciPlus` run exactly replicates the focused 10-edge PAG and broadly supports the other skeletons. It also shows why PAGs require disciplined interpretation: the most tempting bidirected treatment-outcome edge is conditional on complete cases and disappears under a stricter alpha, while some endpoint marks change with variable order or violate chronology.

A separate robust application profile addresses the principal empirical failure found by those audits. Across every cyclic variable order it returns the same PAG in all three panels, preserves the main target adjacencies, and reports no fully directed temporal reversal. It does so by leaving unsupported endpoints unresolved and by spending more CI calls. This is the appropriate robustness trade-off for the application: fewer directional claims, backed by stronger reproducibility over the tested orders. It is not proof of invariance to every permutation.

No algorithm is uniformly “best” on the STAR data. The paper-profile self-FCI⁺ is the fidelity and query-efficiency subject. The robust self-FCI⁺ profile is preferred for this applied interpretation under the predeclared order and temporal audits, although its circles identify less direction. R `pca1g` provides valuable independent replication and gives the most plausible direction for the attrition target edge, yet shares the longitudinal reversal and exhibits endpoint order sensitivity. Standard FCI is less economical but leaves the longitudinal target partly unresolved rather than fully reversing it. Credibility must therefore be assigned claim by claim.

The report therefore rejects two extremes. It is too pessimistic to conclude that causal discovery adds nothing to a randomized study; the attrition and stability analysis reveals useful structure. It is too optimistic to conclude that FCI has rediscovered and fully identified the experiment’s causal model. The correct contribution lies between them: an auditable map of conditional independence constraints, uncertainty, and modelling sensitivity.

13.2 Recommended extensions

1. Reconstruct the randomization strata, estimate formal intention-to-treat effects, and model grade-3 observation explicitly.
2. Develop cluster-aware or multilevel CI tests for students nested within classrooms and schools.
3. Report Monte Carlo uncertainty for the 100-replicate PAG stability frequencies and extend stability summaries to endpoint compatibility.
4. Run separate reading and mathematics analyses and examine prespecified treatment-effect heterogeneity.
5. Evaluate defensible temporal background knowledge separately from the data-only robust profile, without silently forcing preferred arrows.
6. Compare alternative mixed-data CI tests; the completed $k = 2, 3, 4$ grid should remain a bounded sensitivity audit rather than a claim about the unknown true degree.
7. Add an independent exhaustive MAG-equivalence oracle for target-PAG derivation and extend stability reporting to endpoint compatibility.

Reproducibility

A.1 Repository layout

Path	Contents
<code>src/fci_engine/</code>	Reusable FCI and FCI ⁺ package.
<code>tests/</code>	Unit, oracle, integration, typing, and report tests.
<code>case_studies/tennessee_star/download_data.py</code>	Official Harvard Dataverse downloader and shape validation.
<code>case_studies/tennessee_star/study.py</code>	STAR cohort construction, coding, discovery, bootstrap, sensitivity, and summary payload.
<code>case_studies/tennessee_star/report.py</code>	Standalone HTML report renderer.
<code>case_studies/tennessee_star/pcalg_reference.R</code>	Independent CRAN <code>pcalg::fciPlus</code> runner and matched G-square CI test.
<code>case_studies/tennessee_star/pcalg_reference.py</code>	R runner integration, artifact loading, and metadata validation.
<code>case_studies/tennessee_star/output/</code>	Machine-readable PAG, benchmark, bootstrap, sensitivity, contrast, and visual report artifacts.
<code>reports/fci_plus_star_report.tex</code>	This complete LaTeX report.
<code>reports/generate_report_figures.py</code>	Vector-figure generator reading the committed result JSON.
<code>reports/generate_oracle_validation_summary.py</code>	Deterministic generator for the Figure 4(b) oracle and seeded SEM evidence.
<code>reports/data/oracle_validation_summary.csv</code>	Frozen target hash, run settings, metrics, query counts, separator provenance, and execution outcomes for Tables 6.1 and 6.2.
<code>reports/generate_scaling_benchmark.py</code>	Deterministic 5–80 node exact-oracle scaling audit.
<code>reports/data/scaling_benchmark.csv</code>	Query counts, timings, and exact-skeleton checks for both scaling families.

Path	Contents
ADVISOR_MEETING_QA_README.md	Meeting-ready scope, methods, interpretation, limitation, and next-step QA.

A.2 Commands

Listing A.1: Reproduce the STAR application.

```
PYTHONPATH=src python -m \
  case_studies.tennessee_star.run_case_study \
  --benchmark-repeats 3 \
  --bootstraps 100 \
  --descriptive-bootstraps 1000 \
  --n-jobs 4
```

Listing A.2: Refresh the independent R reference and rebuild the integrated application.

```
PYTHONPATH=src python -m \
  case_studies.tennessee_star.run_case_study \
  --refresh-pcalg \
  --rscript /path/to/Rscript \
  --benchmark-repeats 3 \
  --bootstraps 100 \
  --descriptive-bootstraps 1000 \
  --n-jobs 4
```

Listing A.3: Regenerate the report figures.

```
PYTHONPATH=src python reports/generate_report_figures.py
```

Listing A.4: Regenerate the frozen Figure 4(b) validation summary.

```
PYTHONPATH=src python \
  reports/generate_oracle_validation_summary.py
```

Listing A.5: Run the software quality gates.

```
python -m pytest -q
python -m ruff check src tests examples case_studies reports
python -m mypy
python -m mypy case_studies/tennessee_star
python -m build --wheel
```

A.3 Data integrity

The committed official data files are:

- `STAR_Students.tab.gz`, a deterministic gzip of the Dataverse tabular export, SHA-256:

```
2ffa578822eb30fcd6626fa7c5cc734a
721fd05dd522403f0d643f89e891bac8
```

- `starUsersGuide.pdf`, SHA-256:

e51ff1d28d5af28c128196b3a133957

f9f2cd872b1abe348f156022193550130

Key STAR Result Tables

B.1 Primary discovery results

Table B.1: Primary full-data discovery results and temporal audit.

Panel	Algorithm	Samples	Edges	CI tests	Seconds	Temporal flag
Attrition	FCI	5,744	14	1,705	5.437	Grade3 Observed $\rightarrow$ K Achievement
Attrition	Self-FCI ⁺ paper	5,744	17	793	1.788	None
Attrition	Self-FCI ⁺ robust	5,744	17	1,304	2.838	None
Attrition	R-FCI ⁺	5,744	13	1,191	10.394	None
Longitudinal	FCI	2,787	10	960	2.148	None
Longitudinal	Self-FCI ⁺ paper	2,787	13	335	0.337	Grade3 Achievement $\rightarrow$ K Achievement
Longitudinal	Self-FCI ⁺ robust	2,787	12	619	0.562	None
Longitudinal	R-FCI ⁺	2,787	12	475	1.650	Grade3 Achievement $\rightarrow$ K Achievement
Focused	FCI	2,976	10	291	0.322	None
Focused	Self-FCI ⁺ paper	2,976	10	185	0.159	None
Focused	Self-FCI ⁺ robust	2,976	10	334	0.247	None
Focused	R-FCI ⁺	2,976	10	256	0.884	None

B.2 Focused sensitivity results

Table B.2: Presence of the class-assignment/grade-3 adjacency across the full alpha, binning, and FCI+ degree-bound grid. Query ranges cover $k = 2, 3, 4$.

Bins	α	Algorithm	k	Adjacent?	CI tests
3	0.01	FCI	–	No	225
3	0.01	FCI ⁺ paper	2–4	No	146–150
3	0.01	FCI ⁺ robust	2–4	No	217–234
3	0.05	FCI	–	Yes, $K_Class \leftrightarrow G3$	305
3	0.05	FCI ⁺ paper	2–4	Yes, $K_Class \leftrightarrow G3$	203–214
3	0.05	FCI ⁺ robust	2–4	Yes, $K_Class \leftarrow \circ G3$	316–367
4	0.01	FCI	–	No	252
4	0.01	FCI ⁺ paper	2–4	No	152–163
4	0.01	FCI ⁺ robust	2–4	No	223–257
4	0.05	FCI	–	Yes, $K_Class \leftrightarrow G3$	291
4	0.05	FCI ⁺ paper	2–4	Yes, $K_Class \leftrightarrow G3$	174–185
4	0.05	FCI ⁺ robust	2–4	Yes, $K_Class \circ - \circ G3$	290–341

Researcher Interpretation Checklist

Before treating an FCI-family result as substantive causal evidence, the following questions should be answered.

1. What is the estimand: a graph, an ancestry claim, or a causal effect?
2. Which variables are pre-treatment, post-treatment, outcomes, or selection indicators?
3. Does the chosen CI test match the data type and missingness pattern?
4. Are observations independent, or is clustering present?
5. Is the FCI⁺ sparsity bound plausible and sensitivity-tested?
6. Does standard FCI recover the same important skeleton relations?
7. Are adjacencies stable under resampling?
8. Are endpoints stable under alpha, binning, and variable-set changes?
9. Do any directed edges violate known temporal order or randomization?
10. Is a bidirected edge being over-interpreted as proof of a specific latent variable?
11. Are design-based conclusions being kept separate from data-only PAG conclusions?
12. Can every published number be regenerated from committed inputs?

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